

SPACE RIDER USER GUIDE



Prepared by	Space Rider Team
Document Type	TN - Technical Note
Reference	ESA-STC-SR-TN-2018-0002
Issue/Revision	2.0
Date of Issue	06/12/2023
Status	Approved

APPROVAL

Title	Space Rider User Guide		
Issue Number	2	Revision Number	0
Author	Space Rider Team	Date	06/12/2023
Checked By Safety	Lucio Gradoni	Date	06/12/2023
			Digitally signed by Lucio Gradoni Date: 2023.12.07 16:10:08 +01'00'
Checked By Security	Danilo Ingami	Date	06/12/2023
		Danilo Ingami	Digitally signed by Danilo Ingami Date: 2023.12.07 16:53:42 +01'00'
Approved By	Fabio Caramelli	Date	06/12/2023
		Fabio Caramelli	Digitally signed by Fabio Caramelli Date: 2023.12.07 17:18:22 +01'00'
Authorised By	Dante Galli	Date	06/12/2023
			Digitally signed by Dante Galli Date: 2023.12.07 17:22:23 +01'00'

CHANGE LOG

Reason for change	Issue Nr	Revision Number	Date
Update to system CDR status close-out	2.0	0	06/12/2023

CHANGE RECORD

Issue Number	2	Revision Number	0
Reason for change	Date	Pages	Paragraph(s)
Update to system CDR status close-out	06/12/2023	All	All

DISTRIBUTION

Name/Organisational Unit
Releasable to the Public

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1 EXECUTIVE SUMMARY

1.1 Scope

This document intends to give a commercial prospective and an overview of the essential technical data to potential Space Rider's Customers. Information about its general aspects is reported in the current section while system architecture, data regarding the system interfaces, the exposed environments, as well as available features and services are detailed in the following chapters.

Together with reference documents (see section 8.1) the aim of the User Guide is to preliminary assess the compatibility of a P/L and its mission with Space Rider, to constitute the general payload service provisions and specifications, and to initiate the P/L preparation of all technical and operational activities for the launch and its safe return on Earth.

Terms and definitions used in the document can be found below (see section 1.3) as well as acronyms table (see section 9).

The Customer who is willing to explore flying solutions with Space Rider is welcome to contact ESA for further details.

1.2 Responsibilities

This document is produced with the contribution of TAS-I as Flight Segment (FS) entity responsible for the cargo-bay interfaces. Images and illustrations are courtesy of TAS-I and AVIO.

ESA is the end responsible of the SR User Guide for the Maiden Flight.

1.3 Definitions

Multi-Purpose Cargo Bay (MPCB)	The Space Rider cargo-bay, designed to host multiple Payload with different needs.
Compartment	Reference volume inside the Space Rider MPCB, defined by specific interface features with the hosted Payload(s).
Payload (P/L)	Object allocated in the MPCB. It could either be a Laboratory carrying Experiments or an individual Experiment.
Standard Payload (STD)	Payload that does not require late access/early retrieval capability nor field of view or direct space exposure.
Late-Access (LA) Early Retrieval (ER) Payload	Payload that requires to be installed just prior to launch and/or to be recovered just after landing (e.g., environmental sensitive P/Ls).
Payloads Aggregate (PLAG)	The configured list of P/Ls qualified for a specific Space Rider mission.
Aggregate Design Authority (ADA)	The Entity in charge of detailed design, development, I/F control, verification, missionization (i.e., mission timeline definition), integration/de-integration and support to flight operations of the PLAG dedicated to each specific mission.
MPCB Operator¹	The Entity in charge of the service management, definition of the Aggregate configuration, final approval of Aggregate detailed design, negotiation and contract signature with Customers, verification of vehicle services compliance with Customers requirements and P/Ls compliance to applicable mission constraints, management of the End-to-End Payload

¹ Limited to SRS Maiden Flight preparation, this role will be covered by ESA.
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	Aggregate mission, follow-up, and approval of the Aggregate Design Authority activities.
Mission Operator	The Entity(ies) in charge of the launch preparation, the launch execution, and the management of the Spacecraft mission until landing and retrieval.
Customer / Sub Aggregator	The Entity owning one or more Payloads.
Laboratory	A facility dedicated to hosting and managing a defined number of Experiments during the mission.
Experiment	A unique item accommodated in a Laboratory or provided as an individual item.
Experiment Owner	The Entity in charge of the development of an Experiment.
Cartridge	Elements of one or more Experiments that are installed in the Payload through the Late Access service and/or retrieved in Early Retrieval, due to their sensitivity to the waiting time before experiment conditions are met.
Missioning Hardware	Customized hardware to meet aggregate-level requirements, designed by ADA for a specific Space Rider mission.
Extension-Kit	Hardware designed to enhance SR built-in capabilities to meet payload-level requirements upon Customer request.

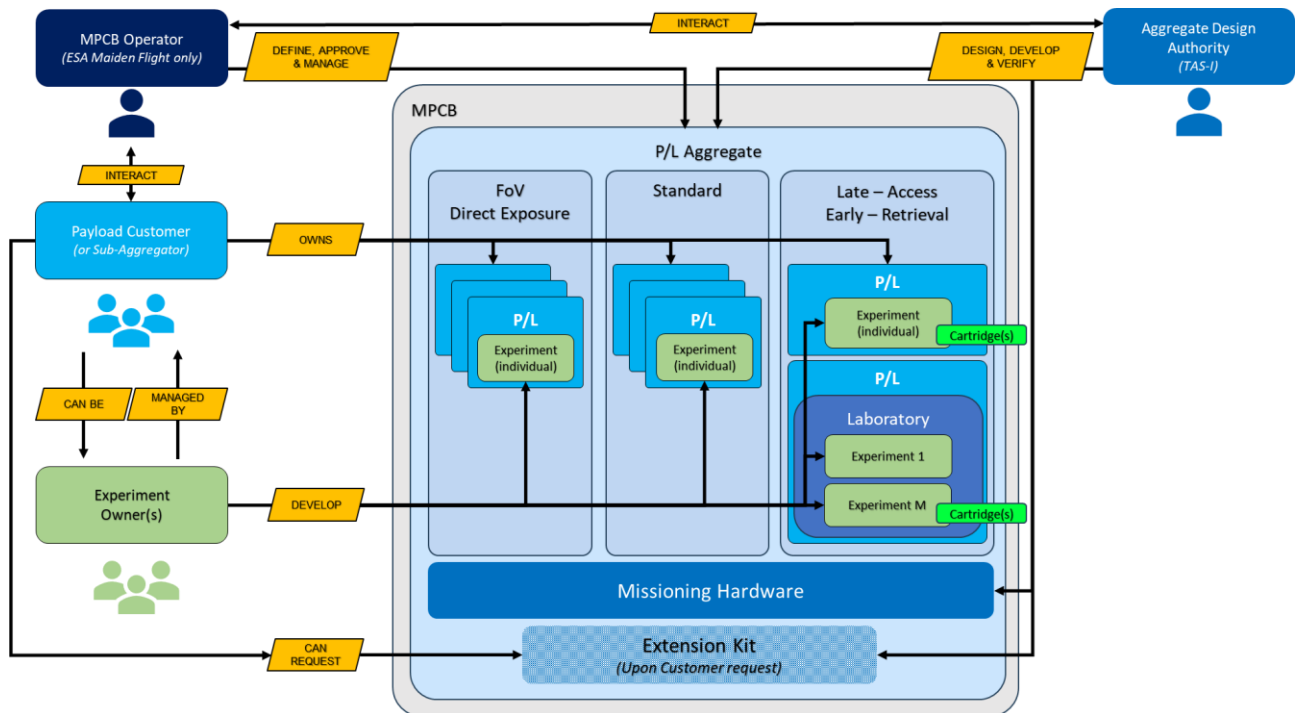


Figure 1-1: Definitions graph

1.4 Introduction to Space Rider System

This section will introduce the Space Rider System describing its composition and general aspects, features and capabilities and a description on why you should consider flying with it.

1.4.1 What is the Space Rider?

The Space Rider System (SRS) is the first European affordable, independent, re-usable, uncrewed end-to-end commercial transportation system for routine access to and return from Low Earth Orbit (LEO). When referring to the SRS, we consider the entire ecosystem of flight segment, ground segment and facilities that are needed to perform a mission and to support the Customers for their business. The Space Rider (SR) vehicle, as the main part of the SRS flight segment, is an uncrewed robotic laboratory about the size of two minivans composed by a re-entry module and an orbital module. Its dynamic configuration allows P/Ls for an array of applications, orbit altitudes and inclinations (compatible with the performance of the launcher), and mission durations. After being launched it will stay in Low Earth Orbit (LEO) for about two months, while performing experiments inside its cargo bay such as technology demonstration and research activities in different fields (e.g., pharmaceuticals, biology, physical science, ...). At the end of its mission, the re-entry module will return to Earth and land on a dedicated area to retrieve the flown payloads while the orbital module performs a destructive re-entry. The re-entry module is then refurbished for another flight.



Figure 1-2: The Space Rider (SR) vehicle.

1.4.2 What does Space Rider offer?

The Space Rider will offer a unique set of capabilities and services to Customer's P/Ls:

- A design based on flight-proven ESA IXV Vehicle.
- Extended exposure (two months or more) in Low Earth Orbit (LEO) environment.
- Open Multi-Purpose Cargo-Bay (MPCB) with a wide Field of View (FoV) for Earth or deep-space observation, and fine-pointing capability.
- Microgravity environment with low g-forces and jitter during flight with high-quality microgravity environment reaching levels down to 5×10^{-6} g.
- Reduced in-flight safety constraints vs human spaceflight safety standards and factors.
- Platform provided in-flight services (i.e., attitude control, power, thermal, telemetry and telecommands) as well as on-ground services (i.e., telemetry stations, in-orbit control centre, user's stations for monitoring of in-flight operations).
- Short time-to-flight, with a maximum advanced flight booking period of one year².
- Reusability roadmap with organisation for a launch rate of up-to 2 launches per year allowing shorter time-to-space and shorter time-to-data.
- Standard payloads installation in the cargo bay, as well as late-access installation at the launch pad until a few hours before launch for environmental sensitive P/Ls.
- Return capability with high down-mass / up-mass ratio and precision soft-landing due to the use of a parafoil that provides governability, both in gliding and in descent speed.
- Early post-landing retrieval of environment sensitive P/Ls with fast access directly at the landing area.
- Post-flight retrieval of Standard Payloads at dedicated Landing site facility.
- User privacy protection at different levels (e.g., experiment design, data, operations, industrial knowledge protection).

² From the FSA signature.

1.4.3 Typical Space Rider P/L applications

The Space Rider provides easy access to and return from space to payloads requiring micro-gravity and/or exposure to the space environment, with accurate pointing capabilities, at lower price and shorter time conditions with respect to other commercially available competitors.

The Space Rider will enable a wide variety of applications such as (but not limited to):

- Microgravity experimentation
- In-orbit Demonstration (IOD) & Validation (IOV) of technologies for exploration, orbital infrastructure servicing, Earth observation, Earth science, Telecom, ...
- In-orbit Services (IOS) for Earth monitoring, satellites inspections, ...
- European pathfinder for commercial services in access and return from space (i.e., commercial space manufacturing).
- Educational missions.

A non-exhaustive list of the scientific and technology research fields that may benefit from the Space Rider services is provided in [ANNEX A].

1.5 Mission Phases and Profiles

This section presents a typical sequence of events for Space Rider mission profiles.



Figure 1-3: Depiction of Space Rider mission phases.

1.5.1 Pre-launch (including Late-Access activities)

This phase includes different activities such as pre-integration and tests, P/Ls accommodation, transport to launch site, final integration and tests, transport to launch pad and integration with launcher, installation of environmental sensitive P/Ls during Late-Access operations through the fairing doors and the side panels of the vehicle.



Figure 1-4: Late-access capability prior to launch.

1.5.2 Launch and ascent

This phase starts from the launch on-top of VEGA-C until the injection of SRS into orbit and the commissioning to begin its orbital phase and operations. Space Rider System operations are autonomous during this phase, and no telecommands will be sent from the Mission Control Centre (MCC) during launch and ascent.

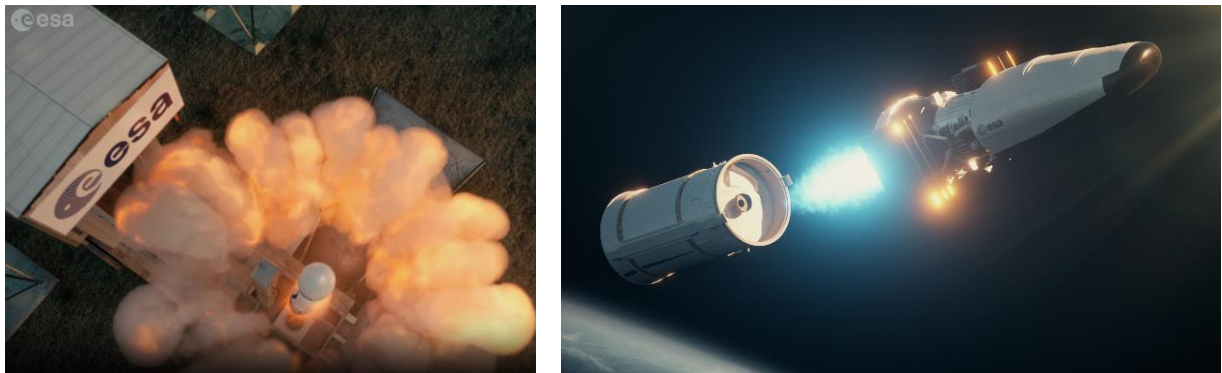


Figure 1-5: Launch and Ascent phases.

1.5.3 Orbital flight (including Commissioning)

This phase starts with the commissioning and vehicle-level preparations to initiate P/L level activities (e.g., solar array deployment and cargo-bay door opening). Payloads' operations follow a mission timeline, managed by the Mission Control Centre (MCC). This period will last two months or more, with each orbit lasting approximately 90 minutes in a typical 400 Km orbit.

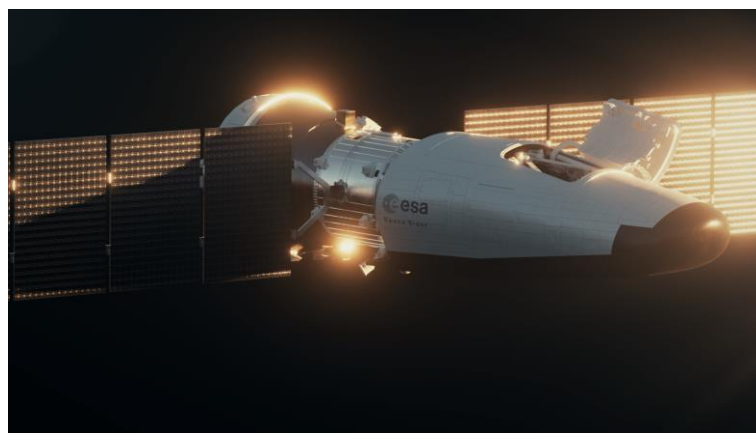


Figure 1-6: Orbital phase.

1.5.3.1 Mission Profile and Orbital Attitude Modes

During the orbital phase P/L Customers will operate for the aim of a specific purpose (e.g., exploit a micro-gravity environment, observation of a specific target, high dissipation, etc.) and this will require that SR to maintain a specific attitude for a specific amount of time. The Payload's needs and relative Space Rider attitude(s) are agreed before the flight, during the mission analysis phase. The below table shows an example of possible SRS attitudes for a quasi-equatorial circular LEO orbit:

Attitude Mode	Primary Pointing	Secondary Pointing	Microgravity Level	Typical P/L
Nose-to-Nadir	SR Nose to Nadir	Cargo-Bay to Anti-Velocity / Velocity Vector (RWs OFF, MTQ ON)	Extreme micro-g 5×10^{-6} g	Biological
		Cargo-Bay to Anti-Velocity / Velocity Vector (RWs ON, MTQ ON, RACS ON for desaturation)	Standard micro-g 5×10^{-5} g	IOD
Tail-to-Sun	SR Tail to Sun direction	-	Standard micro-g 5×10^{-5} g	Biological High dissipation
Nose-out-of-Plane	SR Nose normal to Orbital Plane	- Z RM	Standard micro-g 5×10^{-5} g	Biological
Bay-to-Earth	Cargo-Bay to Nadir	SR Nose to Velocity Vector	Standard micro-g 5×10^{-5} g	Earth Observation
Bay-to-Space	Cargo-Bay to Zenith direction	SR Nose to Velocity Vector	Standard micro-g 5×10^{-5} g	Deep Space Observation
Inertial	MPCB Inertial Fixed	-	Standard micro-g 5×10^{-5} g	Deep Space Observation
Bay-to-Out-of-Plane	Cargo-Bay to Space	SR Nose to Velocity Vector	Standard micro-g 5×10^{-5} g	Deep Space Observation

Table 1-1: Orbital attitudes modes

Nose-to-Nadir (N2N)

Performed pointing the nose of the SR Vehicle to Nadir and assuming the Cargo-bay pointing to velocity vector. This attitude is foreseen mainly for Extreme micro-g services (see 7.5.2).

Tail-to-Sun (T2S)

Performed pointing the tail of the SR Vehicle to Sun direction and the secondary pointing assuming the SR Cargo-bay pointing to orbital plane. This attitude is foreseen mainly for Standard micro-g service and for high dissipation needs (see 7.5.1).

Nose Out-of-Plane (OOP)

Performed pointing the nose of the SR Vehicle normal to orbital plane and the secondary pointing assuming the SR Cargo-bay parallel to orbital plane. This attitude is foreseen mainly for Standard micro-g service (see 7.5.1).

Bay-to-Earth / Nadir (B2E)

Performed pointing the SR Cargo-bay to the direction of Earth (i.e., to Nadir) and the secondary pointing assuming the SR Vehicle nose pointing to velocity vector. This attitude is foreseen mainly for Earth observation services (see 7.6.1).



Figure 1-7 Bay-to-Earth attitude mode.

Bay-to-Space / Zenith (B2Z)

Performed pointing the SR Vehicle Cargo-Bay to Space (i.e., to Zenith) and the SR Vehicle nose to velocity vector. This attitude is foreseen mainly for deep-space observation (see 7.6.2).



Figure 1-8 Bay-to-Space attitude mode.

Inertial

Performed when a specific target pointing is required, mainly for space observation (see 7.6.2).

Bay-out-of-Plane

Performed pointing the SR Vehicle Cargo-Bay to space. This attitude is mainly foreseen for Deep-space observation service as alternative to Bay-to-Zenith attitude mode (see 7.6.2).

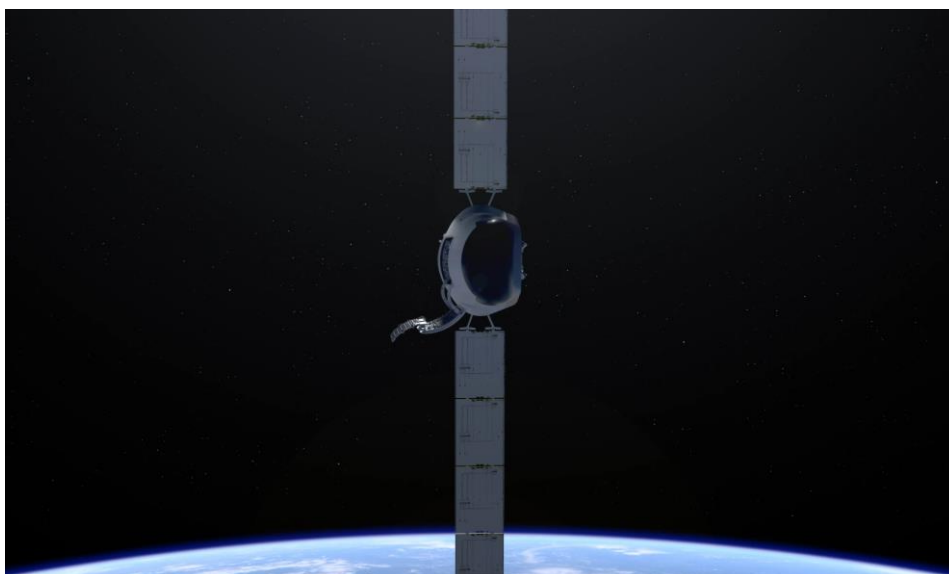


Figure 1-9 Bay-out-of-Plane attitude mode.

1.5.4 Preparation for De-Orbiting

Upon receiving authorization from Mission Control Centre, the Space Rider and its Payloads will be reconfigured for de-orbit (e.g., vehicle avionics cooling-down, radiator stowage, and cargo-bay door closure), waiting for the final authorization for De-orbit that is sent based on the readiness status of all involved parts (vehicle, control centre and landing site conditions).

1.5.5 De-Orbiting

Upon the reconfiguration of the SRS for de-orbit, with activities such as P/Ls cool-down (as preparation for descent environment), radiator stowage, cargo-bay door closure final authorization for de-orbit is given, Space Rider will start an automatic sequence to configure the system for de-orbit and execute the de-orbit boost. This phase ends with the separation of the Re-entry Module from the AVUM orbital one. The re-entry module proceeds to descent phase while the orbital module performs a destructive re-entry.



Figure 1-10: De-orbit, re-entry, and orbital module separation.

1.5.6 Re-entry, Descent and Landing

The Re-entry Module performs autonomously the coasting, re-entry and descent sequences going from hypersonic to transonic flight until the triggering of a subsonic parachute, slowing down at the subsonic regime. At an altitude between 6 and 10 km, there is the deployment of a guided parafoil for a controlled descent up to the Landing Area at the Landing Site, where it reaches ground with proper touchdown conditions and landing accuracy.



Figure 1-11: Re-entry, descent, and landing phases.

1.5.7 Post-Landing (including Early-Retrieval activities)

Upon landing, the re-entry module is monitored, and a remotely guided mobile hangar covers the vehicle prior to operators are allowed to approach and perform the safety activities and early retrieval of payloads. Early-retrieval payloads are handed over to the Customers for processing at a dedicated Payload Processing Facility (PPF) at the Landing Site that guarantees proper level of cleanliness and is equipped with office and laboratory equipment and tools. The re-entry module is moved to another facility for passivation and decontamination of the thrusters, retrieval of standard payloads and preparation for shipment to the refurbishment facilities.



Figure 1-12: Post-landing Early-retrieval phase.

1.5.8 Post-Flight

A post-flight analysis is performed to identify any non-nominal behaviour and refurbishment needs. The re-entry module undergoes detailed inspections of all subsystems to complete the refurbishment sequence and the maintenance or replacement activities are performed. Upon this, the re-entry module is ready for a next flight with a new orbital module, within six-months.

1.6 In-Orbit Services and Close-Proximity Operations Capabilities

Designed to be an affordable, independent, re-usable, uncrewed end-to-end commercial transportation system for routine access to and return from Low Earth Orbit (LEO), its concept can be extended to serve as In-Orbit Services (IOS) and Close-Proximity Operations (CPO) platform in addition to a pure commercial exploitation medium.

Studies for the implementation of technologies and system architecture evolutions are ongoing to support in-orbit operations. Some related features and capabilities are currently present, others will be implemented in the near future or will be designed on-demand to support more complex tasks. In the figure below is shown a vision for Space Rider in-orbit servicing capabilities:

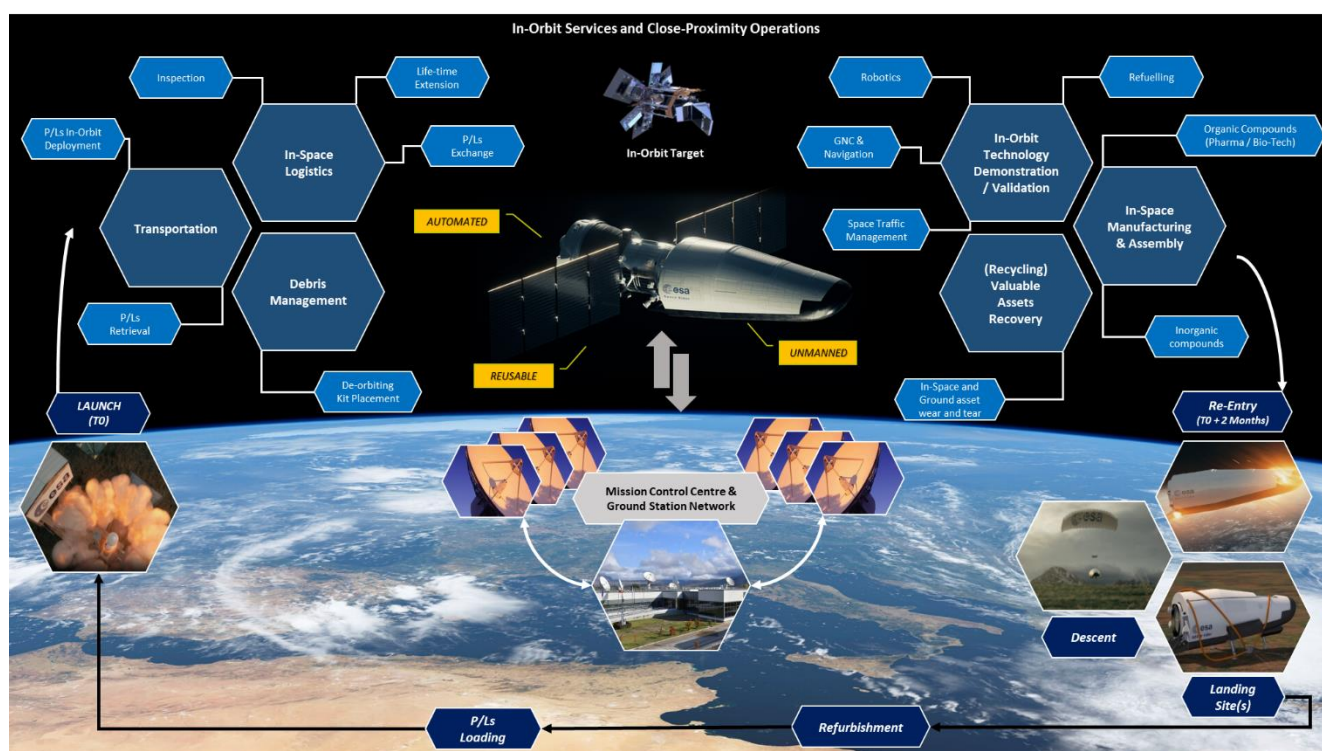


Figure 1-13: Space Rider vision for In-Orbit Services.

1.6.1 Current Capabilities

The SR Vehicle is natively designed to perform IOD/IOV of technologies developed for the In-Orbit Services and Close Proximity Operations domains with the unique capability to return the hosted dedicated P/Ls back to Earth (e.g., for further analyses or possible re-use). In addition, SR can perform orbital manoeuvres and maintain specific attitudes for the release of small satellites or CubeSat from P/Ls dispenser during its mission providing in such a way a transport capability as a service. For this purpose, activities are ongoing to consolidate the definition of proper zones, forbidden areas, and relative approach corridors to the SR Vehicle.

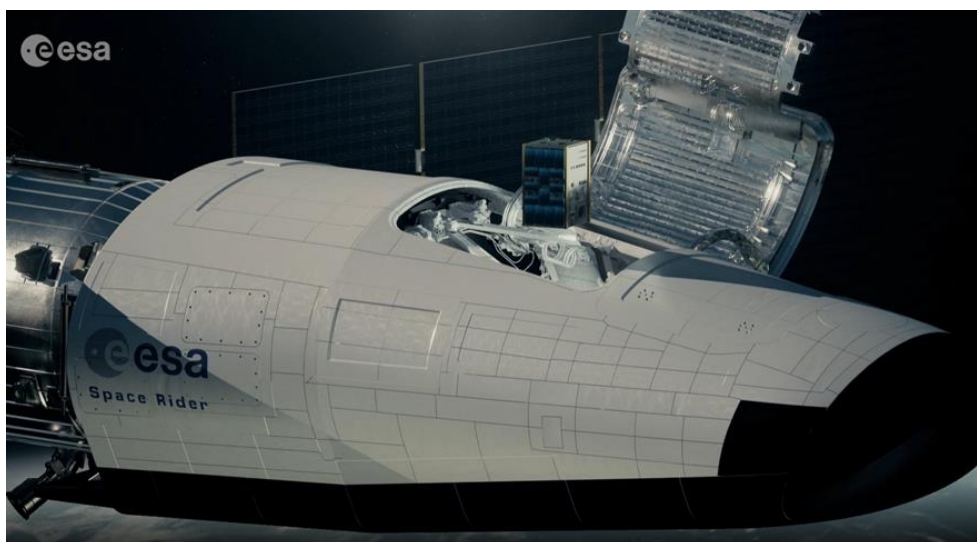


Figure 1-14: Example of P/L separation from SR.

1.6.2 Future Capabilities

The SR Vehicle and its architecture can be enhanced to support advanced IOS/CPO capabilities, making it a cooperative (i.e., capable to control its state in terms of orbit / attitude and communicate with the visiting vehicle) and prepared (i.e., equipped with proper interfaces) platform. For example, to support Close Proximity Operations, studies are ongoing on equip the vehicle with visual markers to support both relative navigation (from rendezvous up to a very-close distance from vehicle) close and capture phases (docking / berthing operations). In addition to allow (passive) docking and/or berthing operations, dedicated analyses are ongoing for the integration of a standard-based mechanical capture interface able to sustain

design loads and rigidity during joint operations. These operations can vary from: operate from/to SR Cargo-Bay exchanging P/Ls with other space infrastructures (e.g., space-tugs), life-extension or de-orbiting tasks, inspection, refurbishment, assembly and/or manufacturing. Many of these activities also requires a robotic arm to perform complex manipulating tasks, and in this domain, activities are ongoing to host a set of robotic arms demonstrators in order to better address the above-mentioned capabilities.



Figure 1-15: Space Rider mission's concept for IOS and CPO – artistic impression.

1.7 Commercial Exploitation

ESA is committed to support and provide access to space to European research, development, and commercial entities. Commercialisation potential and European competitiveness are focus areas for ESA which is mapping the value chain, European supply, and world-wide demand with the final goal to promote international market access to Space Rider that plays a game-changer role in these emerging markets thanks to its flexibility and potential short lead time response to Customer needs. For example, microgravity exposure capability is fundamental for development and commercialisation of new products, dedicated market surveys suggest forthcoming steep increase of market demand for experimentation and in-space manufacturing in LEO for pharmaceutical and bio-tech applications and current and future IOD/IOV and In-Orbit Services / Close Proximity Operations related markets in Europe and worldwide.

2 SPACE RIDER SYSTEM ARCHITECTURE

The Space Rider System (SRS) is composed by the vehicle itself and all the ground-based sites, facilities and equipment used by operators and support personnel for the management of the spacecraft for the launch, in orbit operations, distribution of payload data and telemetry, re-entry, landing and refurbishment phases. The overall SRS architecture is composed by:

- The Flight Segment
- The Ground Segment
- The Landing Site(s)

In addition, the SRS will use and will be interfaced with a Launch System to reach orbit and a Launch Base from which performs its mission.

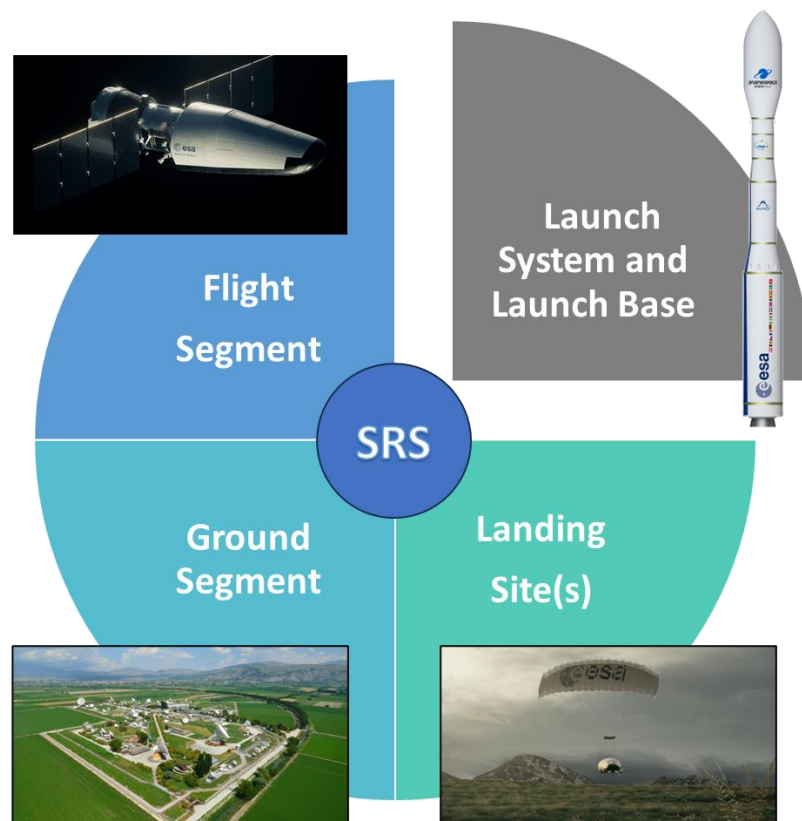


Figure 2-1: Space Rider System (SRS) Architecture

2.1 Flight Segment

The Flight Segment (FS) is the part of the SRS architecture that will comprehends the Space Rider Vehicle, that will reach the orbit carrying the payloads inside its cargo-bay and all the related facilities for integration, test, and refurbishment in addition to all the ground support equipment and test models. In particular, the FS include:

- Space Rider Vehicle:
 - Including its Multi-Purpose Cargo Bay (MPCB), hosting the payloads.
- Integration, Test and Refurbishment Facilities.

In addition, the FS will use and will be interfaced with different Ground Support Equipment and Test Models (e.g., RM Avionic Test Bench or ATB).

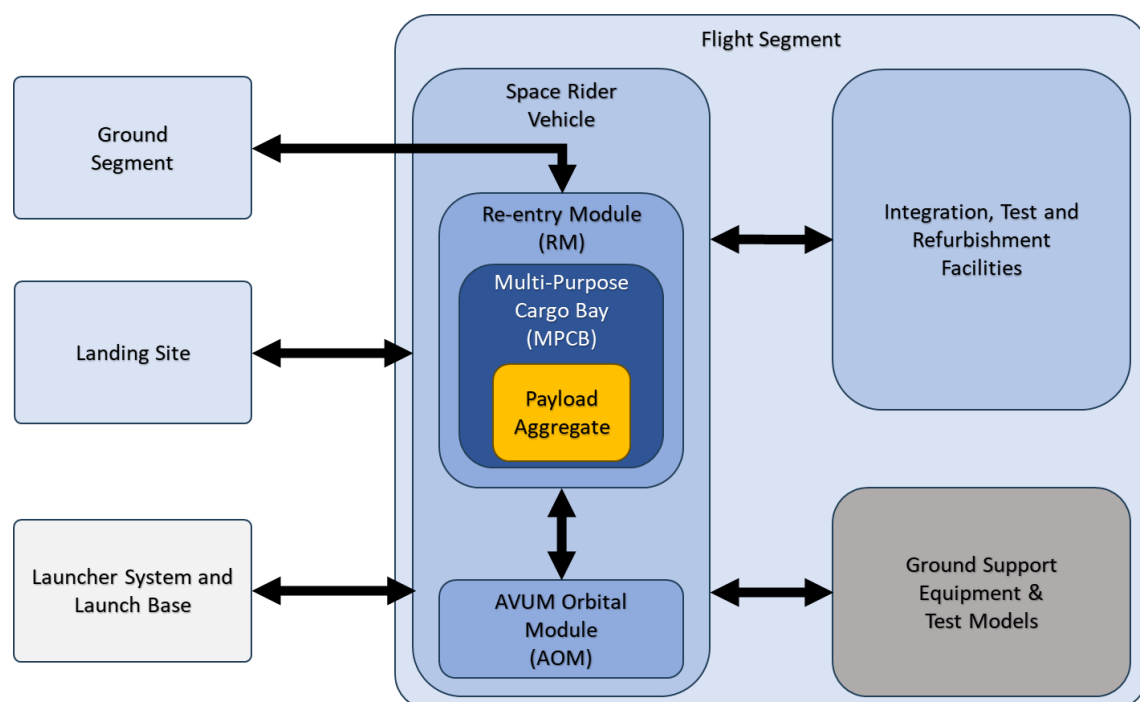


Figure 2-2: SRS Flight Segment architecture.

2.1.1 Space Rider Vehicle

The Space Rider Vehicle is an uncrewed spacecraft, part of the SRS flight segment, composed by two modules, the AVUM Orbital Module (AOM) and the Re-entry Module (RM).

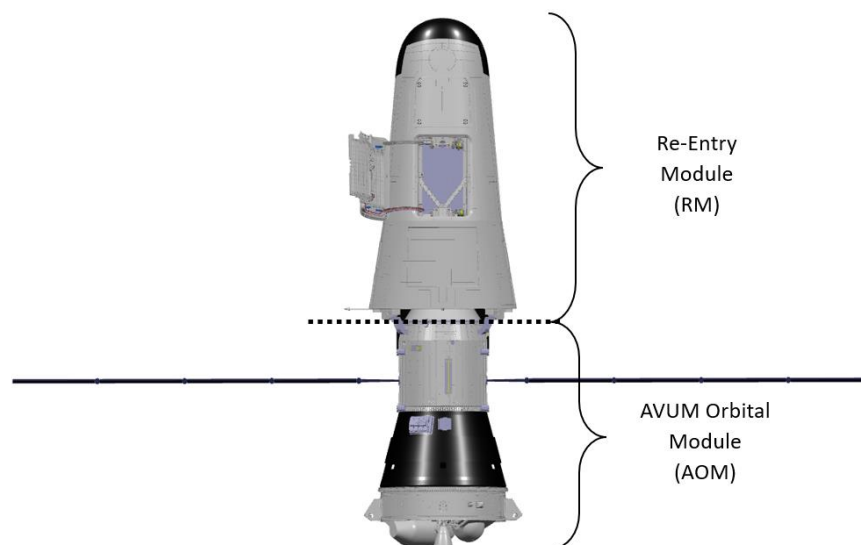


Figure 2-3 Space Rider Vehicle.

AVUM Orbital Module (AOM)

The AOM is a modified version of the VEGA-C upper stage, able to supply power, perform manoeuvres and provide attitude control to the SR vehicle, up to the separation of the two modules. It is made of a slightly adapted VEGA-C upper stage and the ALEK (AVUM Lifetime Extension Kit), a newly developed avionic module whose purpose is to provide electrical power (through dedicate solar arrays, see Figure 2-4) and attitude control. After separation and de-orbiting boost, AOM is disposed by means of a destructive atmospheric re-entry.

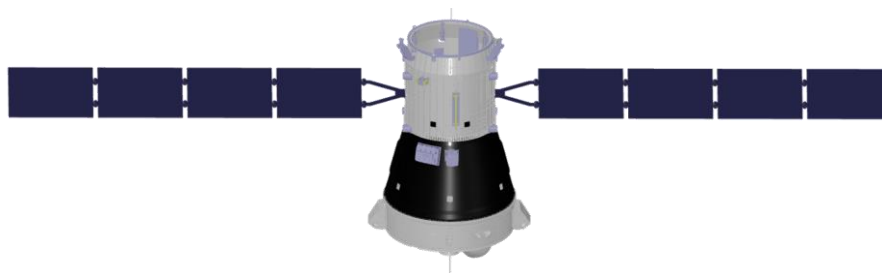


Figure 2-4 AVUM Orbital Module (AOM) artist's impression

Re-entry Module (RM)

The RM is a modified version of the IXV (Intermediate eXperimental Vehicle³) equipped with a dedicated cargo-bay for hosting P/Ls, able to perform ground landing and to re-fly after refurbishment. The cargo-bay door opens during the orbital phase to allow observation and exposure to the space environment, and to provide thermal control via a deployable radiator. The RM is also equipped with two lateral compartments, that can be accessed right before launch and after landing, dedicated to environmental sensitive P/Ls. At the end of its orbital phase, the RM is separated from the AOM, performs atmospheric re-entry and a guided landing on ground where P/Ls are retrieved. The RM is designed to perform 6 flights, upon limited refurbishment.



Figure 2-5 Space Rider - Re-entry Module (RM)

Multi-Purpose Cargo Bay (MPCB)

The P/Ls hosted on a SR flight will be accommodated in the so-called Multi-Purpose Cargo Bay (MPCB) a dedicated section of the RM where P/Ls interfaces and services are provided. The MPCB allows the accommodation of multiple P/L configurations as well as the necessary structures for mechanical fixation and thermal control. It can accommodate sealed or vented P/Ls, directly or partially exposed (e.g., direct illumination, field-of-view, etc.) to space

³ https://www.esa.int/Enabling_Support/Space_Transportation/IXV
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environment. The internal geometrical volume available goes up to 1.2 m³, able to accommodate up to 600 Kg of P/L instruments⁴ mass. The maximum allowed mass for the Payload Aggregate is also driven by launcher capability.

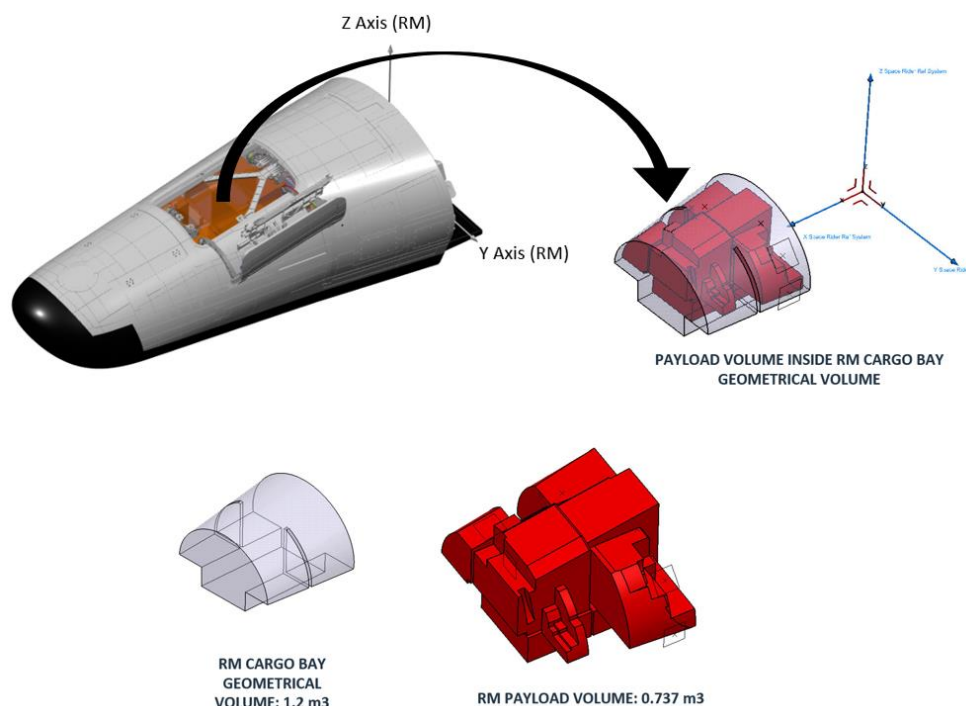


Figure 2-6: The Multi-Purpose Cargo Bay (MPCB)

The accessibility to the MPCB is guaranteed by the MPCB door (as shown in Figure 2-7) that allows an access dimension of 1200 mm x 718 mm (including door fixed mechanisms) during the Standard Payload integration and retrieval operations. For the Late-Access integration and Early-Retrieval operations of environmental sensitive P/Ls, the RM is also equipped with two lateral doors (see Figure 2-7) that allow an access dimension of 575 mm x 262 mm dedicated to access to the outer face of P/Ls mounted on selected LA/ER Support Plates (see section 3.2.1 for further details). A digital PDF-based interactive 3D model of the SR MPCB can be provided on demand for a more detailed overview [RD9].

⁴ Net instruments, including dedicated P/L hardware (e.g., brackets, adapters, etc.) but excluding any missioning hardware required for the instrument to be operated since part of the SRS provided services (e.g., thermal control, telemetry, power, propulsion, attitude, orbit control, etc.).

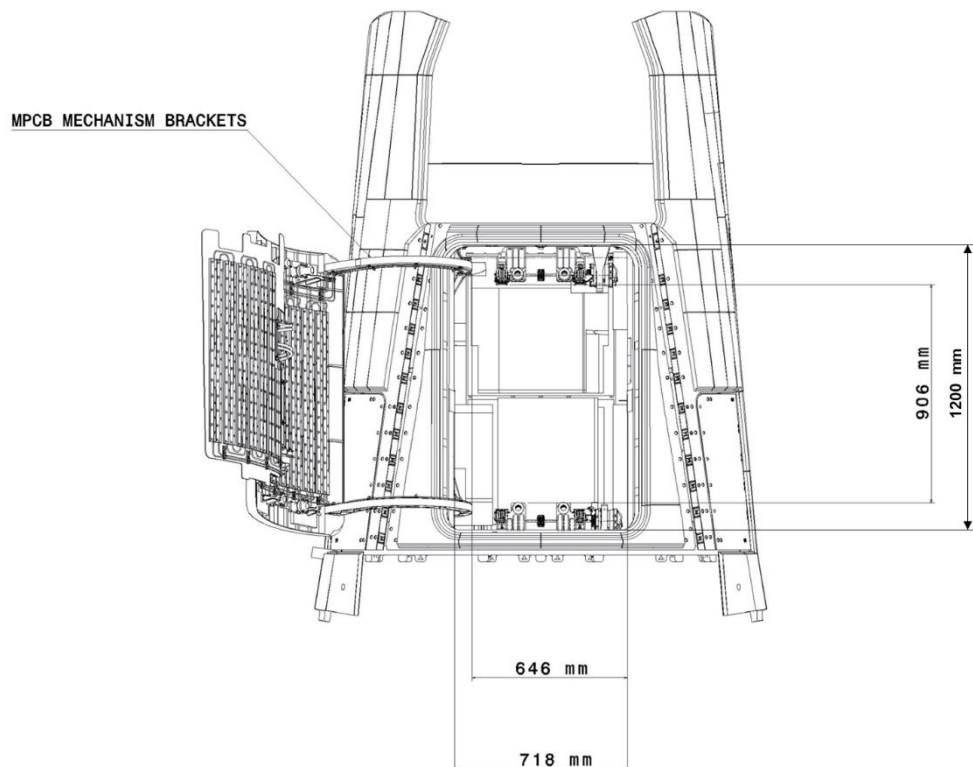


Figure 2-7 MPCB Main door access main dimensions.

The MPCB provides 7 P/Ls aluminium made Support Plates (SP) which purposes are to be:

- The standardized mechanical fixing interface between P/Ls and the RM cold structure.
- The thermal conductive path between the P/Ls and the RM Thermal Control System.

The Figure 2-8 shows the position of the Support Plates within the RM MPCB. For further details on the Support Plates as mechanical and thermal interfaces see section 3.2 and 3.3.

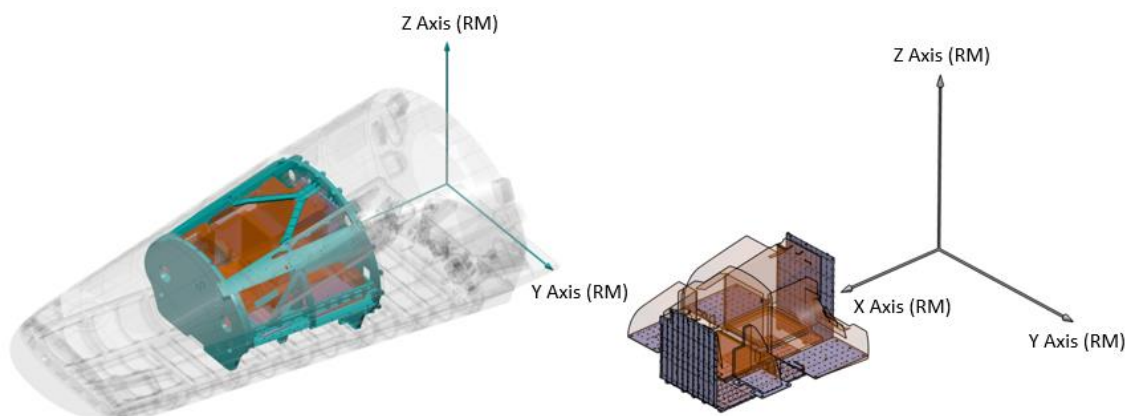


Figure 2-8 MPCB Support Plates overview

Integration, Test and Refurbishment Facilities

For what concerns integration, test, and refurbishment facilities for P/Ls and SR vehicle⁵:

1. The RM will be prepared at Integration Facility at TAS-I premises in Turin (TO), Italy. Here will also be executed the P/Ls fit check on the MPCB of the RM Flight Model (FM) and the preliminary integration tests between P/Ls electrical and functional test models and the RM Avionic Test Bench (ATB) (for further details see section 5.6.4 and 6.4).
2. For the Maiden Flight (MF) only, prior to be moved to the launch site, the SR vehicle will be shipped to the ESA ESTEC in Noordwijk, The Netherlands for its environmental test campaign. After that, at same location, the final integration tests will be performed between the delivered Customer's qualified P/Ls and the RM Flight Model (FM).
3. At this point the SR will be shipped to the launch site at Europe's Spaceport in Kourou, French Guiana. Here remaining P/Ls that are environmental sensitive and or require late-access capability will be lately integrated.
4. After the flight, the P/Ls are recovered and returned to the Customers and the RM will be shipped to the Refurbishment Facility at TAS-I premises in Turin (TO), Italy.

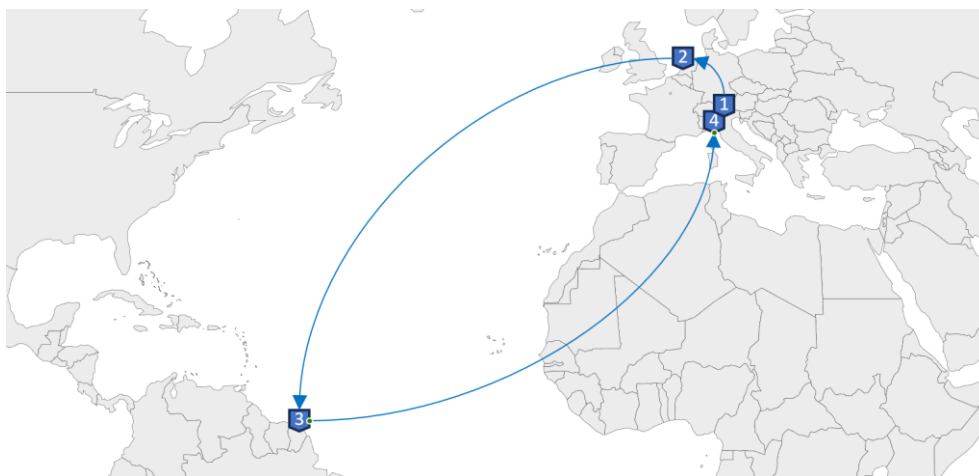


Figure 2-9: Integration, Test and Refurbishment Facilities and locations.

⁵ For a more comprehensive list of P/Ls test and activities, their related locations and time schedule please refer to the ANNEX B.

2.2 Ground Segment

The Ground Segment (GS) is in charge of vehicle and P/Ls monitor and control from target orbit achievement until early post landing phase when the vehicle is switched off. It is composed by all the facilities and equipment based on the ground for the management of the SRS operations and its activities include (but not limited to):

- Receiving and processing TM data both in real time and through TM dumps.
- Prepare, validate, and upload TC products both for S/C and for Payloads as needed.
- Plan and re-plan mission activities including payloads operations.
- Ensure attitude and orbit control as required by system and payloads needs.
- Monitor the AOM during its re-entry phase.
- Monitor the RM during atmospheric re-entry phase as allowed by the duration of the blackout phase and availability of Ground Stations.
- Monitor and command as needed the RM during approach and landing operations.
- Support the RM activities during the post landing phase up to the vehicle switch off.
- Archive data (received TM and sent TC) and generate long term archiving products.

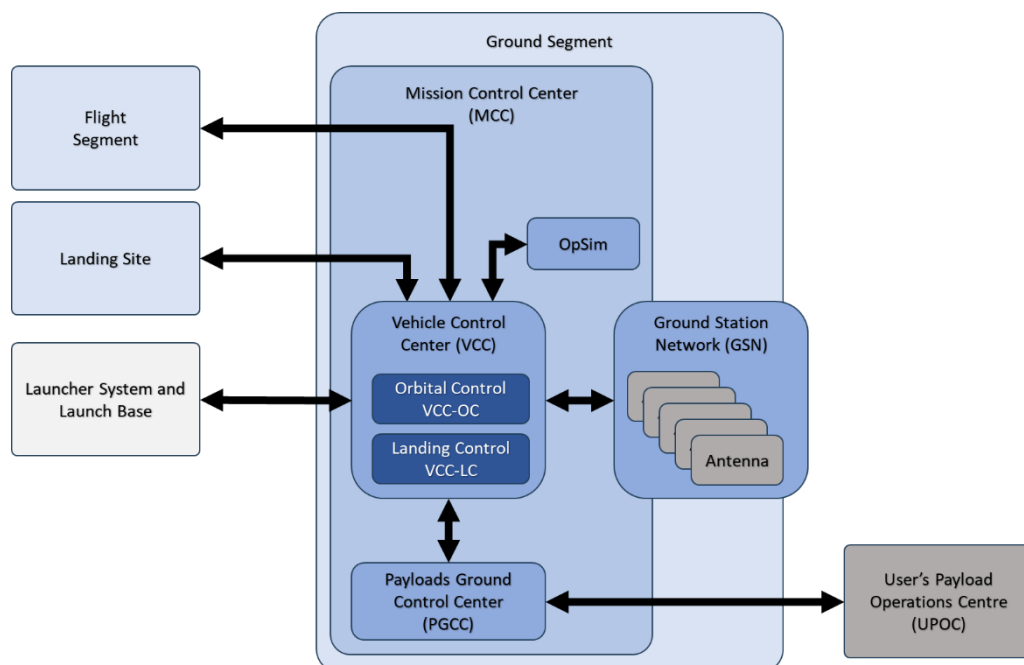


Figure 2-10: SRS Ground Segment architecture.

2.2.1 Mission Control Centre (MCC)

The Mission Control Centre is the part of the G/S dedicated to the control of the vehicle (and its payloads) during the launch, orbital and landing operations. It is composed by:

- The Vehicle Control Centre (VCC) composed by:
 - Vehicle Control Centre – Orbital Control (VCC-OC)
 - Vehicle Control Centre – Landing Control (VCC-LC)
- The Payloads Ground Control Centre (PGCC)

All the communications between the SR vehicle and the MCC during the mission is guaranteed by the use of a world-wide Ground Station Network (GSN).

Vehicle Control Centre (VCC)

The VCC is the part of the Ground Segment that is in charge to control and monitor the SR vehicle during all its mission timeline, sub-divided in orbital and landing control centre.

Vehicle Control Centre – Orbital Control (VCC-OC)

Located at Fucino Space Centre, Italy, under Telespazio (TPZ) responsibility. It is in charge of monitoring and control the SR vehicle after orbit achievement and until de-orbit (including AOM re-entry), exchanging data with Launch Site, worldwide located Ground Stations, and Payloads Ground Control Centre (PGCC) in Turin via secured network links. It is also in charge of organizing and coordinating the Ground Station Network (a mix of ESTRACK and Commercial Ground Stations used to allow world-wide coverage for a SR mission).

Vehicle Control Centre – Landing Control (VCC-LC)

Located in Turin, Italy, under ALTEC responsibility. It is in charge of monitoring and manage the RM during re-entry, descent and landing phases, exchanging data with the Landing Site (also managing the Landing Site Ground Station), ensuring this way the successful recovery of the RM at the end of the mission.

Payloads Ground Control Centre (PGCC)

Located in Turin, Italy, under ALTEC responsibility. It is in charge to manage the P/Ls operations during the orbital phase and to provide to the P/L Customers user dedicated services via secured network link. The PGCC is dedicated to the monitoring and control of the P/Ls, interfacing with the User's Payload Operations Centre (UPOC) for data exchange.

The ground segment, through the PGCC, will make available to the end-users a set of services that will allow an easy management of the P/L requests, their mission planning, monitoring of their activities and P/L data delivery (see section 7.4.4 for further details).

User Payload Operations Centre (UPOC)

During mission operations, the PGCC will interface directly only with the Customers, though their designated User Payload Operations Centre (UPOC). The PGCC represents the access point for UPOC(s) to the SRS mission and P/L(s) data and provide all the means for Customers to both submit user operation requests for P/L planning activities, monitor P/L status and retrieve P/L acquisition data, other than providing visibility to a subset of pre-agreed information regarding the spacecraft that can be useful for P/L planning (see section 7.4.4 for further details).

2.3 Landing Site(s)

The SR re-entry module is capable to perform a controlled re-entry, descent, and touchdown at chosen Landing Site (LDS), depending on mission characteristics and purposes. The compatible landing site must be in the proximity of an airport or a port with a dock and equipped with proper landing areas and facilities to support the P/Ls retrieval operations and the preparation for the shipment and refurbishment phase of the RM. The currently identified Landing Site(s) for the SR vehicle are:

- The Europe's Spaceport in Kourou, French Guiana (for near-equatorial missions and baseline for the MF).
- Santa Maria Airport, Santa Maria, in the Azores, Portugal (for mid-inclination missions).
- An undisclosed location, in Italy.

The identified Landing Site, based on baseline landing site architecture, should provide the following facilities and ground-related elements:

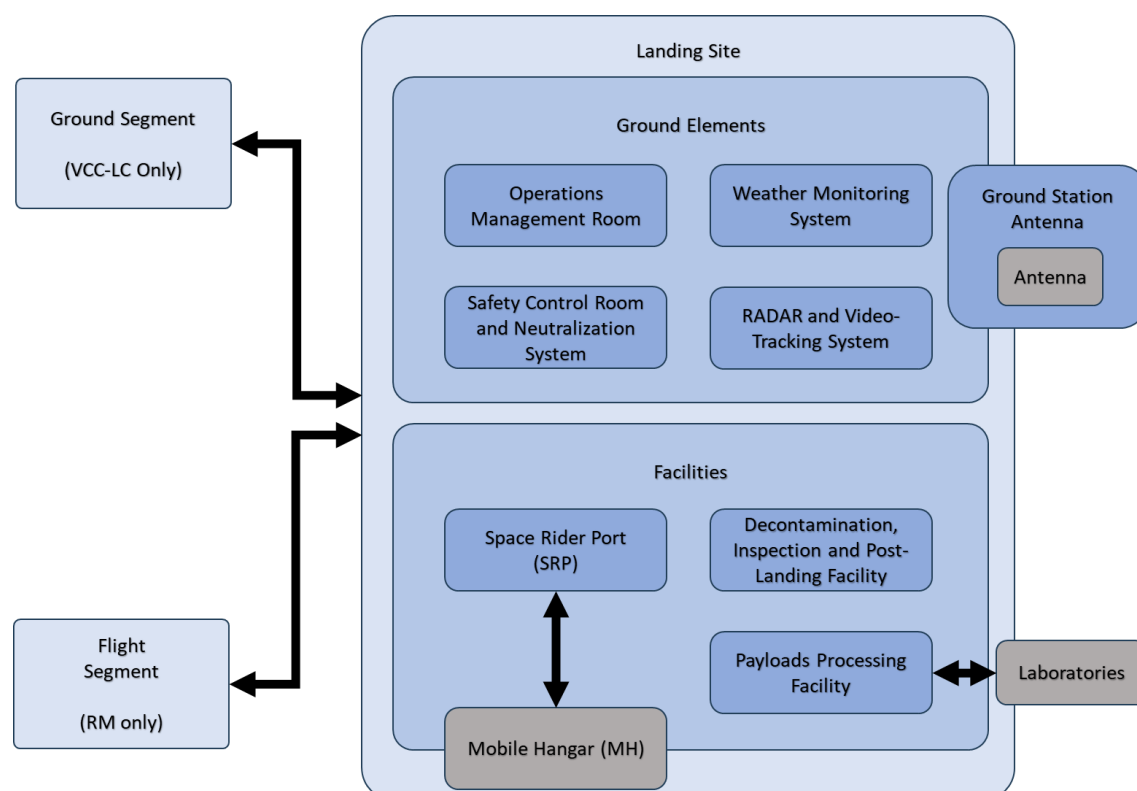


Figure 2-11: SRS Landing Site(s) architecture.

2.3.1 Landing Site Ground Elements

The Landing Site Ground Elements is composed, among others, by the following main blocks:

- Operations Management Room
- Safety Control Room and Neutralization Service
- Weather Monitoring System
- Radar and Video Tracking System

The LDS will also use the Ground Station Antenna to communicate with RM during re-entry.

Operations Management Room

The Operations Management Room hosts the Landing Site Director of Operations (LSDO), Room, who is the responsible for the coordination of all the Landing Site operations. Focal point of the Landing Site in direct voice link with the Operations Director at the VCC. Also reporting the readiness of the Landing Site to proceed with the de-orbiting of the Space Rider.

Safety Control Room and Neutralization Service

Hosts the Landing Site Safety authority, responsible for all the safety aspects of the Landing Site up until Landing, including activating the neutralization system of the RM, if necessary. After landing, the Post Landing Ground Safety Office (PLGSO) takes over the safety aspects.

Weather Monitoring System

It will provide services towards the VCC-LC, including forecast, nowcast, soundings, low and medium altitude wind measurements during preparation, de-orbit, and landing phases.

RADAR and Video Tracking System

The Landing Site will provide both a primary and a secondary RADAR to track the vehicle during the last phase of the mission. The Video Tracking System instead, is capable of visually tracking the vehicle during the final phase of the flight up to 21 km distance and until vehicle touch-down.

2.3.2 Landing Site Facilities

The Landing Site (LDS) Facilities part is composed by four main blocks:

- Space Rider Port (SRP).
- Decontamination Facility, Vehicle Inspection & Check Facility, Post-Landing Processing Facility and Packing, Handling, Storage and Transport Facility.
- Payloads Processing Facilities and Laboratories (PPF & Labs).

At LDS a Mobile Hangar (MH) will also be used to operate on the RM just after the landing.

Space Rider Port (SRP)

The Space Rider Port (SRP) is a dedicated access-controlled area inside the Landing Site (LS) dedicated to RM operations. It comprises the following zones:

- **Landing Area:** dedicated area comprising the Touchdown Zone and the Rollout Zone.
- **Touch-down Zone (TZ):** dedicated area for touch-down of RM inside the Landing Area.
- **Roll-out Zone (ROZ):** dedicated safe zone beyond the Touchdown Zone.
- **Roofed Storage Zone:** dedicated zone with roof close to the Landing Area, to accommodate the Mobile Hangar (MH), RM GSE and offices.

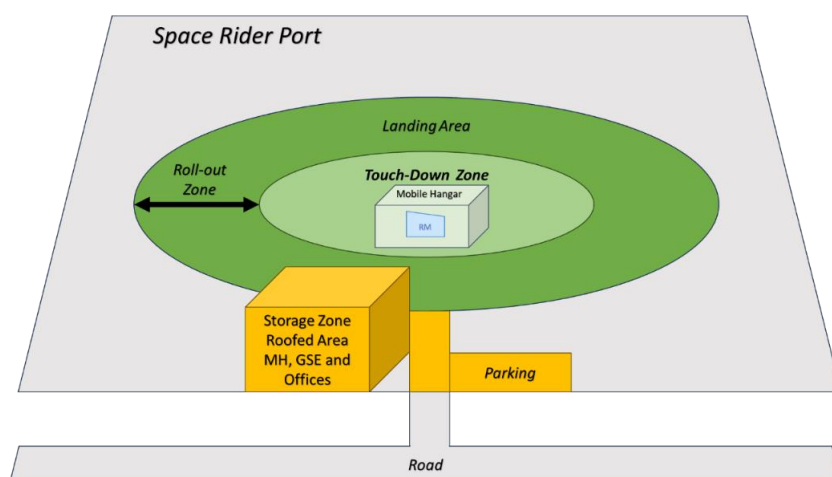


Figure 2-12: Space Rider Port

Before being transported to the Post-Landing Processing Facilities, the RM will undergo cool-down and power-down phases, deactivation of the electric and pyrotechnic circuits and retrieval of environmental sensitive P/Ls directly at the Landing Area with the use of the Mobile

Hangar (MH) equipped with thermal, contamination sensors and an air conditioning system to protect and cool-down the RM immediately after landing. After about an hour, escape suit operators enter the MH and retrieve the environmental sensitive P/Ls via RM side panels.

Facilities for the Management of the Vehicle

The RM will then be moved to the decontamination where all hazardous material remaining will be passivated or decontaminated to keep the subsequent operations on the RM safe. The RM will be inspected, and the operators will proceed to the retrieval of Standard Payloads (consequently moved to the Payloads Processing Facilities), recorded flight and P/Ls data. Finally, the RM will be prepared for storage and shipment to the refurbishment site.

Payloads Processing Facilities and Laboratories

The P/Ls recovered after landing will be moved from the Post-Landing Processing Facility (or directly from the Landing Area for biological sensitive P/Ls) to a dedicated Payload Processing Facility (PPF) via an environmental controlled path. PPF has a size of at least 8 m x 5 m, equipped with regular offices and laboratory tools, refrigerators and one ESD workbench among others. It guarantees ISO 8 level of cleanliness and maintaining a temperature of 22 ± 3 °C and a RH between 45% and 65% (see [RD1] and [RD2] for further details).

- **Laboratories:** Bio-Safety Level-1 class laboratory⁶ will be made available to Customers at the LDS within driving distance from the Landing Area compatible with P/Ls needs.

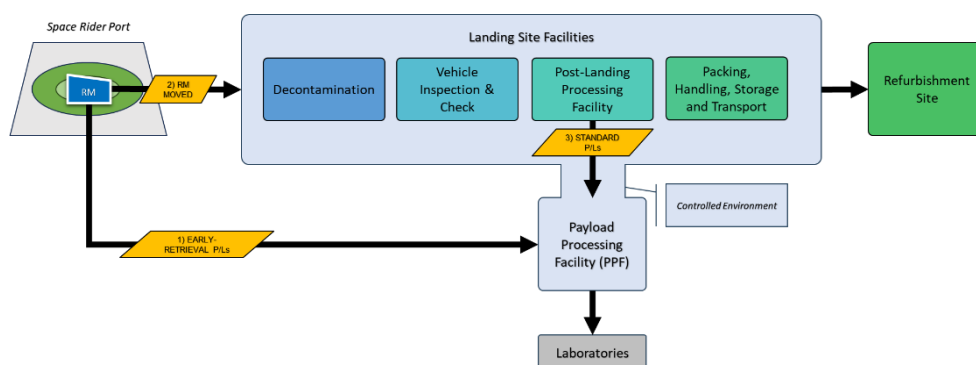


Figure 2-13: Flow of Re-entry Module and Payloads operations after landing.

⁶ Bio-Safety Level-2 (or above) class laboratory availability can be evaluated on demand.
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2.4 Launch System and Launch Site (complementary to SRS)

The Launch System and Launch Site (LS) are respectively the launcher and the spaceport from which SRS will lift-off to start its mission.

- The Launch System
 - VEGA-C launcher
- The Europe's Spaceport in Kourou, French Guiana (including):
 - Launch Site Ground Elements
 - Launch Site Facilities

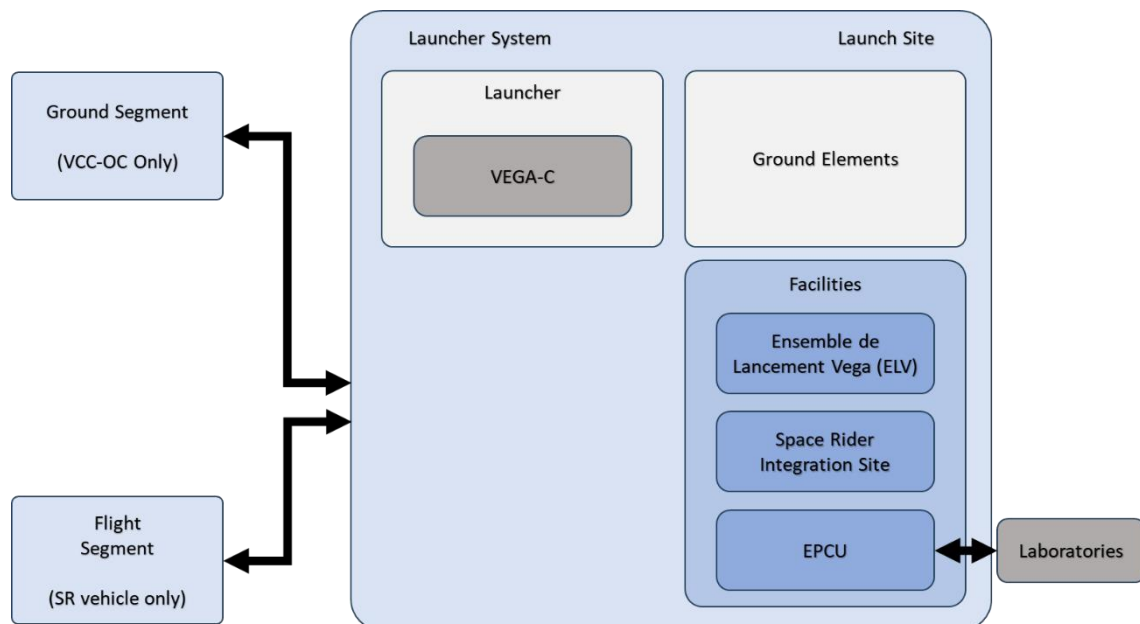


Figure 2-14: SRS Launcher System and Launch Base architecture.

2.4.1 Launcher System

The baseline launch system for the SR is the VEGA-C, an upgraded and more powerful version of the VEGA launcher that increases the performance and the flexibility for multiple payloads missions allowing quasi-equatorial or polar orbits.

A detailed description of the baseline launch system, performance, typical mission profiles, environmental conditions, interfaces and in general all launcher-related aspects can be found in [RD1] and [RD2].

2.4.2 Launch Site Facilities

The Europe's Spaceport in Kourou, French Guiana will be the baseline Launch Site (LS), providing all the facilities and support services to perform the launch of the SR vehicle. The site will also provide the ground segment for the management of launch sequence and early flight phase.

Different facilities and support structures will be made available to the Customers to support late-integration and payload management. A detailed description of CSG facilities, environmental conditions, services (e.g., power supply, communication networks, transportation and handling, fluids, and gases, etc.) in addition to operations policy and planning constraints and in general all launch base related aspects can be found in [RD1] and [RD2].



Figure 2-15: The Europe's Spaceport at Centre Spatial Guyanais (CSG).

3 INTERFACES

3.1 Co-ordinates and Reference Frame

3.1.1 Space Rider Vehicle Reference Frame

The Space Rider vehicle reference frames are a right-handed, orthogonal coordinate systems used for geometrical configuration, design drawings and dimensions.

The Re-entry Module (RM) reference frame has the origin at the base of the RM vehicle, at the geometric centre of the launcher adapter ring, positioned at 206.4 mm long the positive “X” direction w.r.t the interface plane between RM and AOM.

- The positive “X” is along the vehicle centre line positioned at 206.4 mm long the positive “X” direction w.r.t the launcher interface flange between RM and AOM.
- The positive “Y” is, parallel to the side upon which the thermal protection and flaps are, pointing port side.

The positive “Z” is opposite to the thermal protection and flaps. The RM coordinate system is tilted of 1.3 degree about -Y axis with respect to the AOM X axis (the latter parallel to the launcher vertical axis).

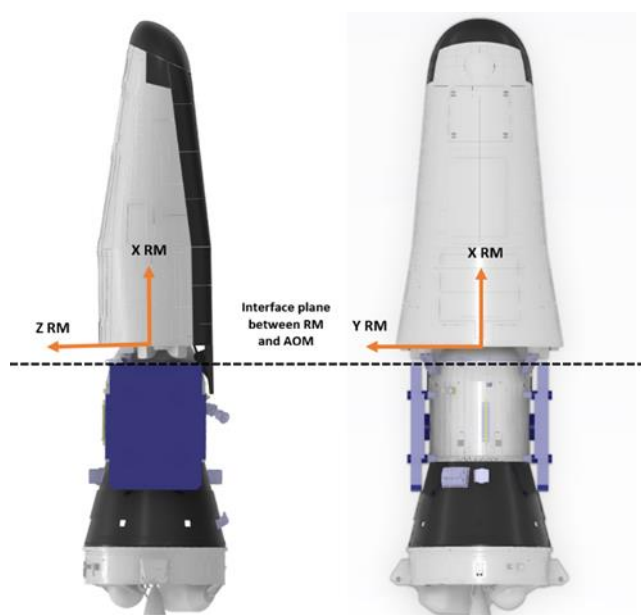


Figure 3-1: SR Reference System.

3.2 Mechanical interface

The MPCB is equipped with so-called Support Plates that provide the mechanical interface to the P/Ls. There are 7 Support Plates, distributed according to the following list:

Support Plate Number	Description / Remarks
Plate #1	Vertical Backward / Aft
Plate #2	Bottom
Plate #3	Vertical Forward / Bow
Plate #4	Late-Access / Early-Retrieval Lateral Starboard (-Y)
Plate #5	Late-Access / Early-Retrieval Lateral Port (+Y)
Plate #6	Lateral Forward Starboard (-Y)
Plate #7	Lateral Forward Port (+Y)

Figure 3-2: MPCB Support Plates definition

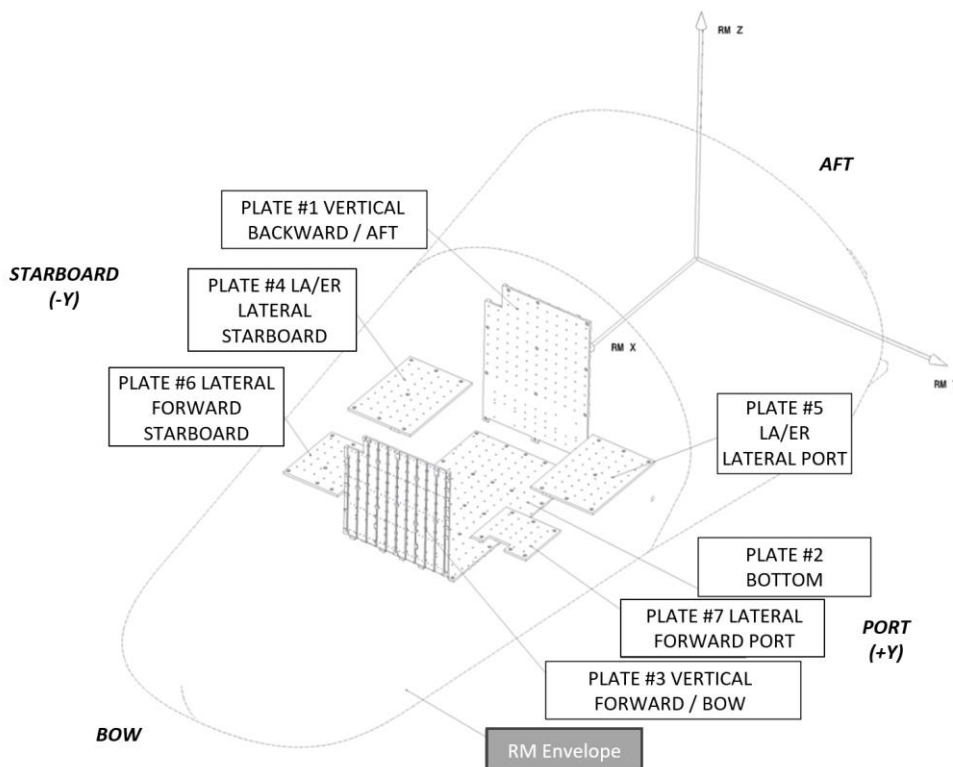


Figure 3-3 MPCB Support Plates allocation

3.2.1 Mechanical interface connections

Standard Payloads plate interface connection

P/Ls installed during standard integration will be hosted in one of the MPCB compartments and mechanical fastening on the MPCB support plates. Each MPCB support plate has the following characteristics:

Parameter	Value
Material	Aluminium
Drilled Holes Dimension	M6 threaded holes
Plate Drilled Pattern	58 mm standard pitch in each In-Plane direction
Thickness	13 mm Lateral plates 8.5 mm Bottom and Vertical Plates
Flatness	<0.1 mm for surface 100x100 mm ² ≤0.2 mm for the overall Support Plate surface
Roughness	< 3.2 µm

Table 3-1: Support Plates mechanical fastening parameters.

The P/Ls mechanical fastening is ensured by the use of fasteners and bolts and the allocation will be iterated and finally determined by the ADA through the PLAG design process after evaluation of P/Ls requirements and aggregate's constraints. Any further needs or mechanical interface adaptation rising during PLAG design process should be addressed to the ADA for evaluation.

Late-Access interface connection

P/Ls with equipment that cannot be installed during standard integration (i.e., experiment's cartridges with environmental sensitive material) can be hosted in one of the two MPCB dedicated lateral compartments in order to be accessed by a TAS-I (RM Prime) human operator briefly prior launch (see ANNEX B). The LA compartments are accessed through dedicated launcher fairing's doors and via SR RM lateral panels (see Figure 3-4):

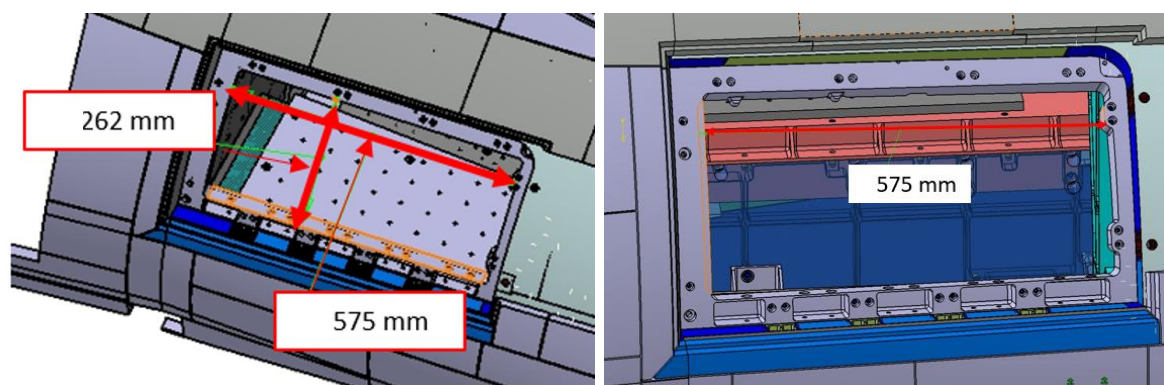


Figure 3-4: RM Late-Access door dimension.

P/L equipment's that requires LA capability shall implement design solutions to guarantee an easy integration flow, intended as an activity that would involve the human operator installing the items (e.g. cartridges) on the P/L docking interface making use of their handling capability, fast or self-locking mechanism in order to ensure required interface connections (e.g.: fastening possibly without the use of tools and/or bolts preferring a plug-in/plug-and-play mechanism). The need of a human operator will impose additional handling, dimension, weight, and safety constraints to LA P/Ls:

Description	Value
VEGA-C Fairing doors diameter	420 mm
RM LA compartment door dimensions	262 mm x 575 mm
LA P/L maximum allowed mass	≤ 6 Kg
Late-Access integration time window	30 min ⁷

Table 3-2: Late-Access interface constraints

⁷ This time is considered for both compartments, operated in parallel (see ANNEX B).

The P/L docking interface part, designed, qualified, and provided by the Customer, that will host the LA equipment(s), will be fixed on the SR RM Support Plates and will provide thermal, power and data connectivity. This will be integrated at the Europe integration site as baseline.

3.2.2 Mass and Volume Capability

The maximum allowed mass for each Support Plate is:

Support Plate	Mass [kg]
Plate #1	65
Plate #2	230
Plate #3	85
Plate #4 Late-Access / Early-Retrieval	70
Plate #5 Late-Access / Early-Retrieval	70
Plate #6	40
Plate #7	40
Total	600

Table 3-3 Support Plates mass capability.

The above reported mass breakdown shall be intended considering eventually P/Ls needed missioning hardware. For the volume capability, due to the irregularity of the MPCB compartments, detailed information about the available volume parameter for each compartment will be provided after P/L preliminary accommodation. A 3D digital model of the MPCB and relative compartments is available and reported as [RD9]. For Deployable-PL and Movable-PL structures the volume and kinematics envelope will further depend on CPO and/or safety constraints.

P/L accommodation in MPCB Support Plates shall be part of PLAG design specific to each mission performed by the ADA. P/L positioning will be subject to this logic in compliance with Customer requirements (e.g., mass, volume, FoV, ...).

3.3 Thermal interface

3.3.1 Thermal Control System

The SR RM can provide thermal dissipation to the hosted P/Ls through the Thermal Control System (TCS) via the MPCB aluminium-made Support Plates which provide the thermal control interface to the P/Ls. The Support Plates provide a heat conductive path to a heat pipe network and a deployable radiator on the MPCB main door (when open) allowing passive heat rejection.

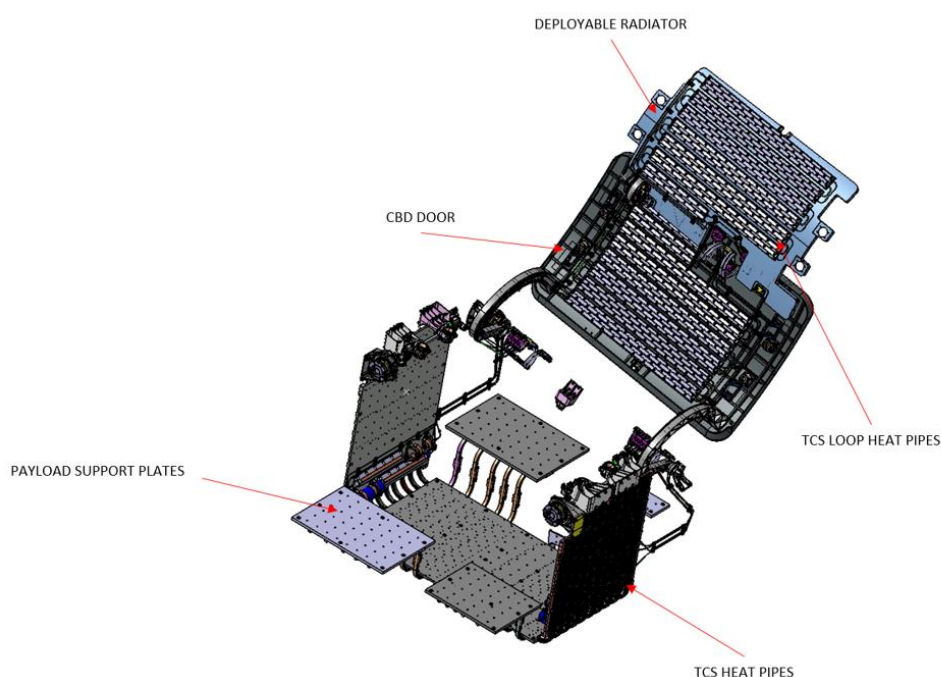


Figure 3-5 RM MPCB Thermal control system

The thermal dissipation capability depends on the current phase of the mission:

Phase	Thermal Dissipation Capability ⁸	I/F Temperature
Pre-launch	50 W limited by MPCB door closed.	$\leq 40 \pm 5 \text{ }^{\circ}\text{C}$
On-orbit	Up to 600 W (attitude dependant)	$[-35 \div 40] \pm 5 \text{ }^{\circ}\text{C}$
	Up to 500 W (attitude dependant, micro-gravity)	$[15 \div 35] \pm 5 \text{ }^{\circ}\text{C}$
Post-Landing	50 W limited by MPCB door closed.	$\leq 40 \pm 5 \text{ }^{\circ}\text{C}$

Table 3-4: Thermal dissipation parameters.

⁸ For the whole Payload Aggregate.

On ground, launch and ascent phases Space Rider can guarantee 50 W of thermal dissipation to the aggregate continuously, this value is driven from thermal dissipation constraints, the cargo bay door is closed, and the thermal dissipation is achieved through conduction of the support plates through the structure of the spacecraft. The aggregate power request will be provided from Space Rider internal battery until the solar array deployment and the value shall not overcome the value of 50 W until the Cargo Bay door will be open and the commissioning phase will be ended.

3.3.2 Thermal Conductive Coupling

Thermal dissipation of the heat generated by the P/Ls on the MPCB Support Plates is ensured by the use of a thermal filler in order to have a good linear contact between the P/L baseplate and Support Plate. The need of any additional thermal provisions required by the P/L, such as thermal spacers or specific thermal filler, shall be notified to the ADA.

The allowable payload base contact heat-flux to MPCB support plate I/F shall be in the range:

Parameter	Value [W/m ²]
Minimum heat flux	0
Maximum heat flux	2000

Table 3-5 Thermal heat-flux range

3.3.3 Thermal Radiative Decoupling

Payloads must be radiatively decoupled from the external environment for their thermal control. Except for the face in contact with the MPCB Support Plate and openings for Payload specific needs, all Payload faces shall be covered in MLI (provided by the Customers). MLI characteristics will be provided upon request.

In addition, the top of MPCB will be closed by a special MLI thermal blanket, acting as sunshade, to prevent sun trapping and solar multiple reflections within the MPCB enclosure. This thermal blanket will be tailored to meet Payload needs related to FoV, visibility for In Orbit Services (IOS) and Close Proximity Operations (CPO).

3.4 Electrical interface

3.4.1 Electrical Power System

The SR RM can provide electrical power supply to the MPCB hosted payloads through the Power Distribution Unit (PDU). The power is routed among the MPCB compartments through lines and the amount depends on the current phase of the mission:

Phase	Power Supply Capability
Other	50 W ⁹ limited by thermal constraints.
Orbital nominal	Up to 600 W ⁹ depending on solar panel orientation and vehicle attitude.
Orbital maximum	A maximum peak of power consumption of 1 kW ¹² can be supported for limited time length ¹⁰ (to be agreed upon optimization).

Table 3-6: Power supply parameters.

During Late-Access operations the power supply will be switched off for 30 minutes for integration. P/Ls shall be able to survive these time windows without any power. Alternatively, they shall include in their architecture an autonomous battery.

During the ascent phase the Payload Aggregate's power request will be met using Space Rider's internal battery until the solar array deployment. The power value during this phase shall not exceed the value of 50 W until the Cargo Bay door is open and the commissioning phase has ended.

3.4.2 Connectors and power lines

Power supply between SR-RM PDU and P/Ls is provided by the use of power lines and connectors from lateral side of the MPCB mid-bulkhead.

⁹ For the whole Payload Aggregate.

¹⁰ During short periods of maximum 90 minutes, once per day for the whole aggregate. During this period the RM attitude-related thermal dissipation capability (see 3.3.1) cannot be overcome.

The overall MPCB electrical power is shared through a total of 11 LCL protected power lines, nominally according to the distribution in the table below (definitive configuration can be different for each mission):

Line nominal power [W]	Number of lines	LCL class	Nominal Current [A]	Output current limit (I_{TRIP})
40	5	2	≤ 2	$2.4 \pm 5\%$
100	3	4	≤ 4	$4.8 \pm 5\%$
200	2	8	≤ 8	$9.6 \pm 5\%$
400	1	16	≤ 16	$19.2 \pm 5\%$

Table 3-7: Power lines typologies.

The maximum peak of consumption of 1 kW can be supported through a combination of the available lines, depending on the aggregate combined needs.

The aggregate power lines will be provided as missioning hardware and manufactured at the proper length to guarantee the capability to provide each Payload with the requested power connection.

Power connectors from RM PDU are installed on a plate in the lateral side of the MPCB mid-bulkhead (see Figure 3-6). Depending on power needs, P/Ls should include in their design the connector(s) of the following typologies:

Line nominal power [W]	Connector Type
40	DEMA-09P-NMB
100	340105601B0X ¹¹ -15-19PN
200	340105601B0X ¹⁰ -13-98PN
400	340105601B0X ¹⁰ -15-19PC

Table 3-8: Power connector types.

The needed connector and its relative position (i.e., which face of the Payload to be installed) shall be iterated and agreed upon with the Aggregate Design Authority.

¹¹ Where 0X is the connector shell type and can be 00 or 07.
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3.4.3 Voltage, current and grounding parameters

Parameter	Value
Voltage	28V unregulated [33 ÷ 26] V
Current	2 – 16A
Grounding	Payload(s) shall provide a single point of grounding (SPG)

Table 3-9: Voltage, current and grounding parameters.

At switch-on and for any mode change, the in-rush charge (current x time) to any P/L shall be limited to the following parameter:

Parameter	Value
In-rush charge	$I_{peak} < 30A$

Table 3-10: In-rush current.

The maximum total charge will be:

LCL Class	Min Tripp off time [ms]	Maximum Total charge [mC]
2	8	16
4	8	32
8	5	40
16	5	80

Table 3-11: Maximum total charge.

The aggregate-level test will be performed considering the maximum and the minimum voltage bus to the P/Ls (compliant to ECSS-E-ST-20-07C).

P/Ls shall make the necessary provisions to have their design compatible with SR power supply.

3.5 Data and Communication interface

The SR RM can provide to the MPCB hosted payloads the capability to send/receive data (TM) and telecommands (TC) through the Mass Memory Unit (MMU) via On-Board Computer (OBC). This information is routed among the MPCB compartments through data lines.

3.5.1 Data and Communications interface on-board

3.5.1.1 Connectors and data lines

Communication between SR-RM MMU and P/Ls is provided by the use of data lines from MMU to P/L data interface connector(s).

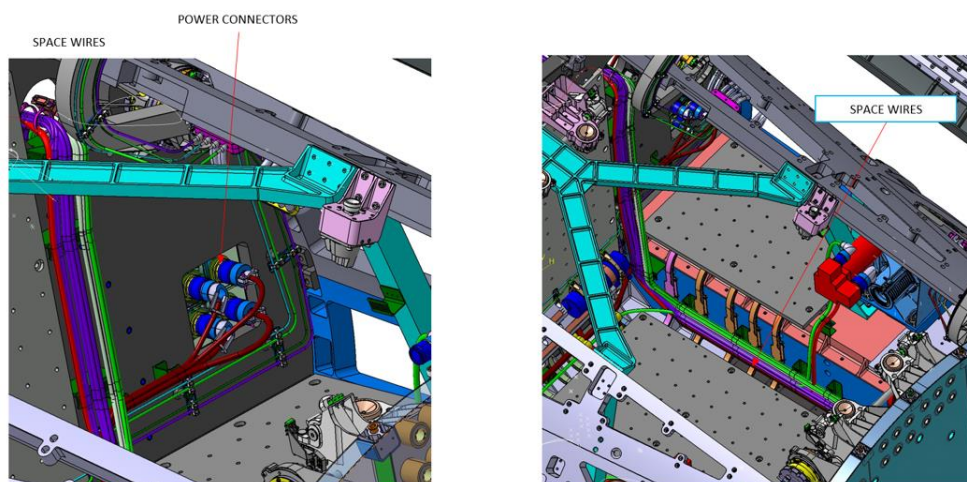


Figure 3-6: Data lines in the MPCB.

Data and communication lines available at MPCB level are:

Line type	Number of available lines
Ethernet	20
SpaceWire	10
RS422	10

Table 3-12: Data and communication line types.

The aggregate data lines will be provided as missioning hardware and manufactured at the proper length to guarantee the capability to provide each Payload with the requested data connection.

The SR-RM data connector baseline is:

Parameter	Value(s)
Data connector types	MDMA-9S

Table 3-13: Data and communication connector types.

The connector's relative position (i.e., which face of the Payload to be installed) shall be iterated and agreed upon with the Aggregate Design Authority.

3.5.1.2 Protocols

Data communication is implemented with different protocols at different levels. The Payloads supported communication protocols are:

Layer	Protocol(s)
Link-level	• Ethernet • SpaceWire • RS-422
Network-level	• IP (Ethernet)
Transport-level	• UDP (Ethernet)
Application-level	• Compliant to the ECSS PUS standard [see RD6], opportunely tailored for SR. • Not compliant to ECSS PUS standard.

Table 3-14: Data and communication supported protocols.

3.5.1.3 Data Rate Transmission On-Board Capability

Data-rate transmission capability depends on the selected link-level protocol. We can have:

Protocol	Data-rate
Ethernet	12.75 MBytes/s to be shared among all the Payloads connected to this interface and transmitting at the same time
SpaceWire	20 MBytes/s to be shared among all the Payloads connected to this interface and transmitting at the same time. Video data from P/Ls to MMU up to 200 Mbps ¹²

RS-422 | 0.125 MBytes/s per payload and therefore constant regardless of the number of payloads connected to this interface

Table 3-15: Data and communication on-board parameters.

3.5.2 P/L Data and Communications interface with ground

The data and communication systems allow the P/Ls to communicate with the MCC through the RM MMU in order to send telemetry and receive telecommands for their operations. The communication link is performed through the use of the S-band.

Parameter	Value
Downlink capability	2 GB / day ¹²
Downlink data-rate	2 Mbps
Storage capability (on-board)	3.6 GB / day ¹²
Uplink capability	1 batch of commands (600 kbit) / orbit ¹²
Uplink data-rate	4 kbit/s

Table 3-16: Data and communication with ground parameters.

3.6 EMC interface

3.6.1 Single point of grounding

Each P/L must provide a single point of grounding interface with respect to MPCB interface.

Each P/L MLI must be grounded with respect to P/L itself.

3.6.1.1 Late-Access Payloads Grounding

Late-Access P/Ls or LA elements must be grounded too, e.g., via bonding strap or blind mate connectors.

¹² For the whole Payload Aggregate.

4 ENVIRONMENTS

During the preparation and integration of the payloads for a launch, in flight, performing in orbit operations and finally during the re-entry, descent and landing phases the spacecraft is exposed to a variety of mechanical, thermal, electromagnetic, and atmospheric-related environments. This chapter provides a description of the different environments that the spacecraft, and its hosted payloads, are intended to withstand during its operational cycle.

4.1 Mechanical Environment

The mechanical environment is described in the present section, and it is exposed showing different aspects of mechanical constraints and loads to withstand during integration, launch and re-entry phases.

4.1.1 Transportation and Handling

During transportation to the Launch Site, post-flight transportation and during integration, the P/Ls shall sustain any simultaneous combination of the Design Limit Loads:

Load Case	Operation	Load factors [g]		
		X	Y	Z
1	Hoisting	± 1.4	± 1.4	± 1.4
2	Handling	± 1.4	± 1.4	± 1.4
3	Transportation	± 3.0	± 3.0	± 3.0

Table 4-1: Transportation and Handling loads.

4.1.2 Stiffness requirements

The MPCB hosted payloads shall maintain the first main mode above the specified level to avoid interactions with SR and VEGA-C launcher structural dynamics:

Frequency type	Frequency range [Hz]
First main mode	> 140

Table 4-2 Primary lateral and longitudinal frequency limit.

In case the above stiffness requirement cannot be met, the P/L Customer shall iterate with the Aggregate Design Authority (ADA) to assess the compatibility with the SR.

4.1.3 Quasi-Static (QSL) and Low Frequency (LFL) Loads

Quasi-static and low-frequency loads are provided to define maximum mounting interface loads that P/Ls could experience during the mission. The QSL is the acceleration applied at the P/L CoG, representative of the force's summation predicted at the Support Plate-P/L interface. During any phase, including launch, the MPCB will not induce to the P/Ls loads from any simultaneous combination, in excess of the following QSL Qualification Loads:

QSL Qualification Loads [g]		
Payload Mass ¹³	IP and OOP [g]	Notes
M ≥ 50 Kg	15	The loads in each direction shall be combined and applied simultaneously
M < 50 Kg	22	

Table 4-3: QSL Qualification loads.

The QSL loads are intended as qualification loads (K_q included) therefore K_p and K_m shall be applied on top to derive the P/L Design Limit Loads according to RD5.

4.1.4 Sine Loads

The Sine loads defined in the table below shall be applied at P/L interface along the three axes:

Sine Spectra – ALL PLATES ¹⁴			
Payload Mass < 50 Kg		Payload Mass ≥ 50 Kg	
Frequency [Hz]	IP and OOP [g]	Frequency [Hz]	IP and OOP [g]
5 – 22	10 [mm] (0 to peak)	5 – 19	10 [mm] (0 to peak)
22 – 110	20	19 – 110	14

Table 4-4 Sine loads.

¹³ The payload mass considered as reference is the nominal one (basic mass).

¹⁴ Spectrum to be applied at the equipment interface.

The Sine loads are intended as limit loads (K_q excluded) therefore K_q , K_p and K_m shall be applied on top to derive the P/L Design Limit Loads according to RD5.

4.1.5 Random Vibro-Acoustic Environment

The P/L shall be able to withstand the random acceleration levels due to acoustic vibrations, Out of Plane and In Plane, in the PSD spectrum below:

Random Vibroacoustic Spectra					
ALL PLATES ¹⁵					
Payload Mass < 50 Kg			Payload Mass ≥ 50 Kg		
OOP			OOP		
Hz	Acceptance PSD [g^2/Hz]	Qualification PSD [g^2/Hz]	Hz	Acceptance PSD [g^2/Hz]	Qualification PSD [g^2/Hz]
20	+ 3 dB/Oct	+ 3 dB/Oct	20	3 dB/Oct	3 dB/Oct
80	0.25	0.5	80	0.15	0.3
500	0.25	0.5	500	0.15	0.3
2000	-6 dB/Oct	-6 dB/Oct	2000	-6 dB/Oct	-6 dB/Oct
g_{RMS}	14.4	20.4	g_{RMS}	11.2	15.8
IP			IP		
Hz	Acceptance PSD [g^2/Hz]	Qualification PSD [g^2/Hz]	Hz	Acceptance PSD [g^2/Hz]	Qualification PSD [g^2/Hz]
20	4 dB/Oct	4 dB/Oct	20	4 dB/Oct	4 dB/Oct
80	0.05	0.1	80	0.05	0.1
1000	0.05	0.1	1000	0.05	0.1
2000	-3 dB/Oct	-3 dB/Oct	2000	-3 dB/Oct	-3 dB/Oct
g_{RMS}	9.1	12.8	g_{RMS}	9.1	12.8

Table 4-5: Random vibroacoustic loads.

¹⁵ Spectrum to be applied at the equipment interface.

The qualification of the Payload could be also performed using the vibroacoustic environment (provided by Space Rider) upon specific request. Random vibration environments from transportation and ground processing are enveloped by the above-specified levels.

The qualification and acceptance random spectra include K_q and K_a respectively, K_p and K_m shall be applied on top to derive the P/L Design Limit Loads according to [RD5].

4.1.6 Shock Environment

This section describes the envelope for the maximum shock levels at the MPCB interface for the hosted payloads (hard-mounted, all-locations). Payloads should maintain structural integrity (no fractures, disassembly, or low-releasable mass) when submitted to the shock environment represented below:

Frequency [Hz]	SRS Shock Response Spectrum [g]
100	40
300	300
1000	1000
10000	1000

Table 4-6: SRS Shock response spectrum.

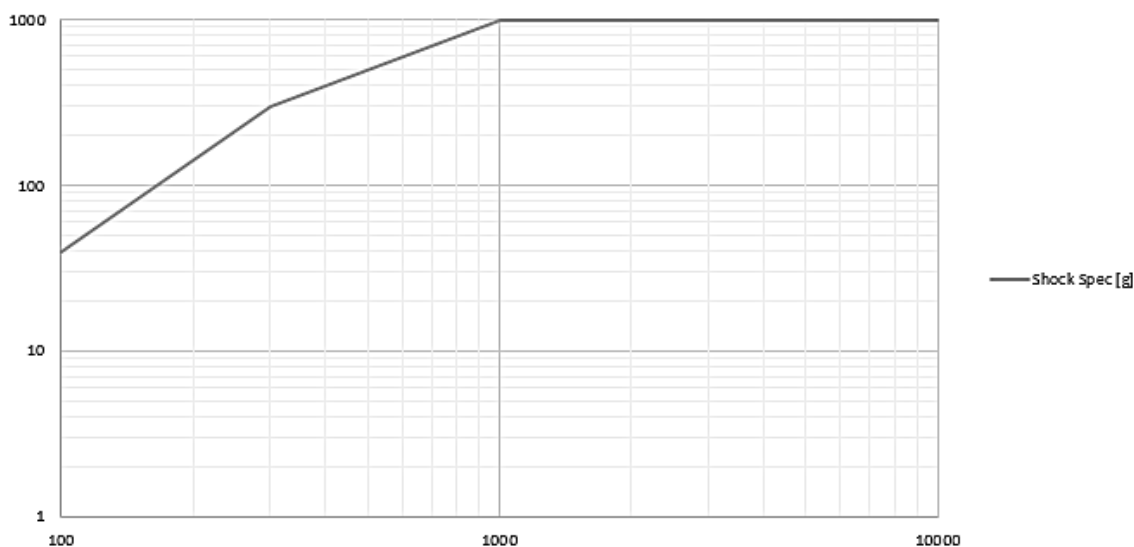


Figure 4-1 Shock level envelope

4.1.7 Pressure Environment

During the mission the SR MPCB is subject to the depressurization during the ascent phase and the re-pressurization during the descent phase. The SR MPCB hosted P/Ls shall withstand the maximum pressure change rates described in the following sub-sections. The P/L design shall assure that any venting for the depressurization and re-pressurization can take place unimpededly and in accordance with the profiles of the MPCB described below.

De-pressurization profile (Launch and Ascent Phase)

During the launch and ascent phase, the SR MPCB environment depressurization rate will be:

Mission Phase	Depressurization rate
Launch and Ascent	$\leq 2200 \text{ Pa/sec}$

Table 4-7 Launch and Ascent depressurization rate.

Re-pressurization profile (Re-entry and Descent Phase)

During re-entry and descent phases until landing, the P/L shall sustain the following pressurization profile:

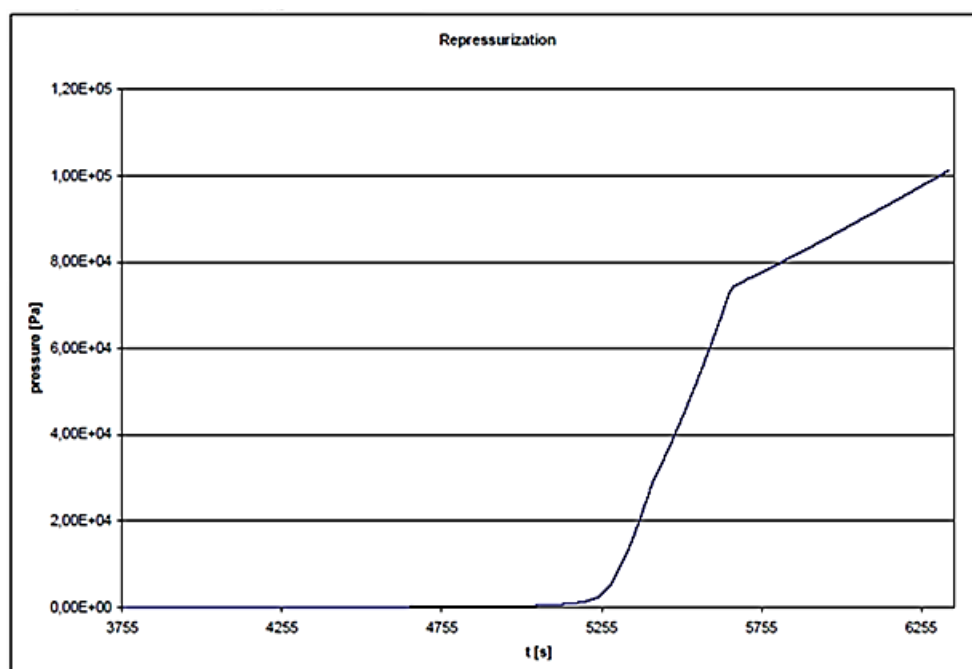


Figure 4-2 Re-pressurization profile.

4.1.8 Microgravity

The microgravity environment, at the P/L interface, during the orbital phase, in any possible attitude configuration and operation (e.g., tail-to-sun) shall respect the following envelope:

- for frequencies (f) $0.01 \leq f \leq 0.1$ Hz, the Root Mean Square amplitude of the residual disturbances will be less than 2×10^{-6} g;
- for $0.1 < f \leq 100$ Hz, the disturbances will be less than the product of $[2 \times 10^{-5} \text{ (g)} \cdot \text{frequency (Hz)}]$;
- for $100 < f \leq 300$ Hz, the amplitude of the disturbances will not exceed 2×10^{-3} g;
- for frequencies (f) above 300 Hz, disturbances with amplitude exceeding 2×10^{-3} g will be indicated on a time scale, in function of their occurrence during a typical reference mission of the Space Rider. In this case, the mechanical subsystem is normally isolated.

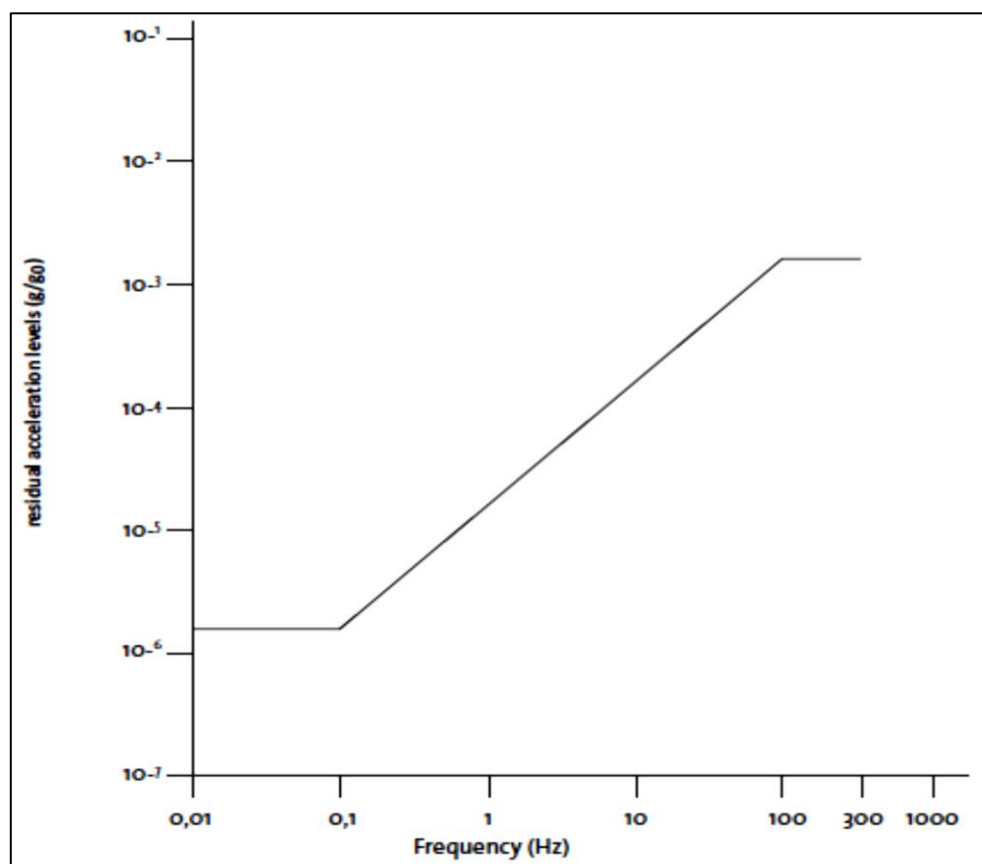


Figure 4-3 Micro-gravity environment at MPCB level.

The above microgravity environment will be guaranteed to P/Ls during orbital phase activities for a specific amount of time according to P/Ls Aggregate design and system constraints.

4.1.9 Re-entry, Descent and Landing loads

During the landing phase (ground impact), the P/L shall withstand the following maximum static acceleration (design limit level) at the RM CoG:

Location	Acceleration [g] ALL AXES
RM CoG ¹⁶	$\pm 8.3^{17}$

Table 4-8: Landing loads.

¹⁶ P/L adjusted or updated values will be determined after accommodation study. Iteration with the ADA to reduce impacts can be evaluated.

¹⁷ The specified value can be updated but will remain enveloped by the expected loads at launch phase.
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4.2 Thermal Environment

4.2.1 Integration, Transportation and Ground Launch Facility Operations

P/Ls that will be integrated in the RM MPCB during AIT activities in Europe will be delivered by the Customers and integrated under the following thermal environmental conditions:

Parameter	Value
Temperature	$22 \pm 3 \text{ }^{\circ}\text{C}$
Relative Humidity	45 – 65 %

Table 4-9: Thermal environmental conditions at AIT site(s) in Europe

After the AIT activities in Europe, the P/Ls will be transported inside the RM placed in its transport and storage container for road / sea and / or air transportation phases from the integration to launch facilities and encapsulation of the SR stack up, through the SR stack mating to the launch vehicle and the stand-by period until launch maintained at the following thermal environmental conditions (see [RD1] and [RD2] for further details):

RM Location / Parameter	Transfer between Assembly / Integration Buildings	On Assembly / Integration Buildings		Transfer to Launch Site	On Launch Pad
Status	Transport Container	Not Encapsulated	Encapsulated (fairing)	Encapsulated (fairing)	Encapsulated (fairing)
Temperature	$[-10 \div 50] \pm 1 \text{ }^{\circ}\text{C}$	$22 \pm 3^{\circ}\text{C}$	Air inlet $[11 \div 25] \pm 1^{\circ}\text{C}$	$\geq 15 \text{ }^{\circ}\text{C}$	Air inlet $[11 \div 25] \pm 1^{\circ}\text{C}$
Relative Humidity	20 – 60 % ¹⁸	55 % $\pm 10\%$	55 % $\pm 5\%$	$\leq 20\%$	$\leq 20\%$

Table 4-10: Thermal environmental conditions during transportation and integration at Launch Site.

¹⁸ With no condensation.

4.2.2 Launch, Ascent and Orbital phases

During the launch and ascent phases the thermal environment is affected by limited dissipation (the MPCB door is closed, and the radiator is not operational, see Table 3-4). After commissioning (i.e., when the MPCB door is open, and the radiator is operational) the P/Ls shall be able to withstand the following environment:

Parameter	Value
External sink temperature	4 K (worst cold case assessment)
MPCB enclosure emissivity	0.8
Solar Electromagnetic Radiation / Solar Particle Radiation ¹⁹	at 1 AU: 1366.1 W/m ² at aphelion: 1321.6 W/m ² at perihelion: 1412.9 W/m ²
Integrated spectral irradiance ²⁰	1366.1 W/m ²

Table 4-11: Thermal environment ascent parameters.

4.2.3 Re-entry, Descent and Landing phases

From the moment the MPCB main door closes, and the radiator stop working, the thermal environment is affected by limited dissipation (see Table 3-4). In addition, we have:

Parameter	Value
P/Ls cool-down phase before de-orbit temperature at the I/F	Down to 5°C
MPCB temperature at Support Plate interface with the P/Ls	≤ 40 °C

Table 4-12: Thermal environment re-entry parameters.

¹⁹ Electromagnetic radiation from the Sun that falls on a unit area of surface normal to the line from the Sun, per unit time, outside the atmosphere at specified distance.

²⁰ The integrated spectral irradiance has been made to conform to the value of the solar constant accepted by the space community.

4.3 Electromagnetic Environment

The electromagnetic (EMC) environment is described in the present section, and it is exposed showing different aspects of EMC compatibility constraints.

4.3.1 RF Compatibility levels

A P/L shall emit, to the antenna terminal, a RF power level lower than 60 dB W with respect to the power level at the fundamental frequency, in the following frequency bands:

- S-band: 2025-2100 MHz
- C-band: 4200-4400 MHz
- C-band: 5400-5900 MHz

This requirement is applicable only for P/Ls with RF transmission equipment. The MIL-STD-461E and CE106 test setup shall be used as guideline for EMC test.

4.3.2 Aggregate Radiated Susceptibility

The P/L Aggregate shall not transmit to the RM, an electromagnetic radiated emission above the following mask:

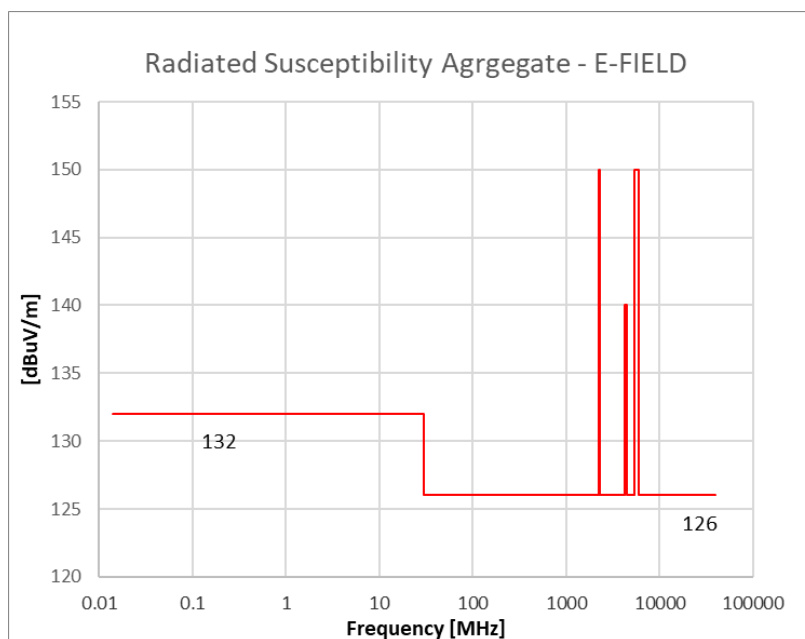


Figure 4-4 Aggregate Radiated Susceptibility.

Frequency Range	Level [dB μ V/m]
14 [kHz] – 30 [MHz]	132
30MHz – 40 [GHz]	126
2200 – 2290 [MHz]	150
4200 – 4400 [MHz]	140
5400 – 5900 [MHz]	150

Table 4-13: Aggregate Radiated Susceptibility levels.

4.3.3 Aggregate Radiated Emission

The P/L Aggregate shall be compatible with the following mask:

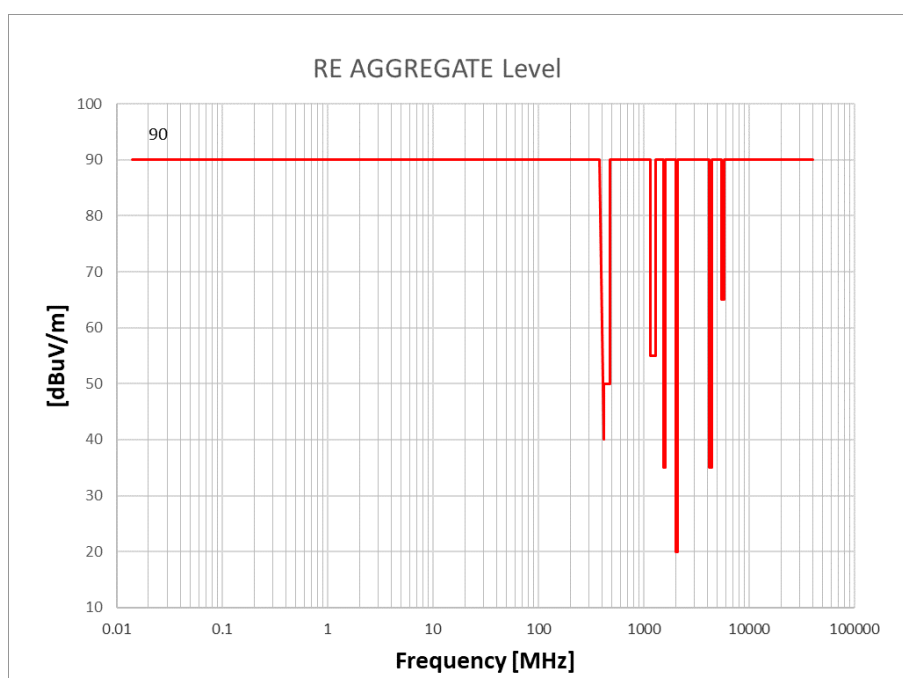


Figure 4-5 Aggregate Radiated Emission.

Frequency Range [MHz]	Level [dB μ V/m]	Notes
417-422	40	Only if the unit is operative during the launcher phase (SE=20dB – Closed door)

420-480	50	Only if the unit is operative during the launcher phase (SE=20dB)
1164-1300	55	Only if the unit is operative during the launcher phase (SE=20dB)
1555-1595	35	GPS
2025-2110	20	S Band
4200-4400	35	Only if the unit is operative during the re-entry phase
5400-5900	65	Only if the unit is operative during the launcher and/or re-entry phases

Table 4-14: Aggregate Radiated Susceptibility.

In case of P/L needs not matching with the above shown notching scheme, further iterations to accommodate EMC compatibility will be possible.

4.4 Space Environment

During in-space operations, the RM MPCB hosted payloads will be exposed to different factors, the main ones are listed below:

- External Vacuum
- Atomic Oxygen
- Cosmic Radiation
- Solar Light

4.4.1 External Vacuum

The space vacuum pressure in orbit is a function of the orbit and the quote. At the altitude for SR reference missions, the pressure of the space environment will be near perfect vacuum:

Mission Phase	Vacuum pressure range
In-Orbit	$< 1 \times 10^{-7}$ Pa

Table 4-15 Vacuum pressure range.

4.4.2 Atomic Oxygen

The P/L shall be designed to withstand a fluency of AO/cm² without degrading their performances as reported below.

Mission Phase	ATOX fluency range
In-Orbit ²¹	1.6×10^{21} Atoms of Oxygen (AO)/cm ²

Table 4-16: ATOX fluency range.

4.4.3 Cosmic Radiation

Maximum external radiation levels on payloads mounted in the opening of the MPCB doors depend strongly on the SR attitude and orbital characteristics. A typical radiation levels during orbital phase at reference altitude can reach up to:

²¹ Worst case scenario in equatorial orbit for 60 days (exposed P/L).
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Radiation type	Level	Note
Solar Flare(s)	Up to 50 rad (0.5 Gy)	Possible during the whole mission
Trapped Particles Exposure	10 rad/day	in case of 400 km polar orbit and considering 1 mm of Al shielding

Table 4-17: Typical radiation levels.

4.4.4 Solar Light

Irradiation of solar light on payloads mounted in the opening of the MPCB doors depends on the spacecraft attitude and orbital characteristics.

Users shall specify their experiments needs in terms of solar constant hours (SCh) per day.

4.5 Cleanliness and Contamination Control

4.5.1 Cleanliness

Payload integration activities will be performed in Europe integration site as baseline in dedicated ISO 8 clean rooms providing air cleanliness via constant positive pressure, careful material selection, all personnel wearing PPE and air filtration and dilution.

Cleanliness at Integration

The MPCB will ensure towards the P/L the following cleanliness levels, during the integration phase:

- ISO 8 class contained atmosphere facilities;
- At integration: PAC [ppm] and MOC [g/cm²] values are currently under evaluation and will be provided at later stage.

Cleanliness for In-Orbit

The MPCB will ensure as a baseline towards the P/L the following cleanliness levels during in-orbit:

- PAC [ppm] and MOC [g/cm²] values are currently under evaluation and will be provided at later stage.

Cleanliness at Post-landing

Landing site design finalisation is ongoing, cleanliness levels at post-landing phase will be provided at later stage.

4.5.2 Contamination analysis

It is recommended to perform a contamination sensitivity analysis at payload level in order to determine areas sensitive or not to contamination. Depending on the results of the analysis, it is recommended to introduce mitigations measures to avoid contamination to sensitive areas. ESA is available and can be contacted for any advice on this topic.

5 PAYLOAD DESIGN, DEVELOPMENT AND VERIFICATION

The P/L design, development and verification process will be performed and managed by the Customers. The P/L should provide relevant documentation to demonstrate compliance to SRS I/F constraints and safety requirements.

5.1 P/L Interface design aspects

During the P/L design process a comprehensive list of aspects must be taken into consideration at different level of detail (depending on the phase of the development). In general, preliminary aspects includes (non-exhaustive list):

- Definition of P/L category (see 5.1.1 for further details), overall dimensions and mass.
- Main elements of the P/L (e.g., Single experiment or Laboratory + Experiments, in the latter case list by broad lines the mission and purpose of all carried experiments).
- Description of P/L thermal control, power and data architecture and budgets.
- Description of any vibration-related (e.g., pumps, etc.) or EMC disturbance (e.g., antenna, electromagnetic actuators, etc.) sources.
- Identification and assessment of safety aspects.
- Operational and ground-segment related operations.

The P/L shall also present a development plan, a description of the P/L mission phases from integration on the vehicle until retrieval after landing (including required P/L technical and schedule conditions and constraints) and a chronology of P/L activities in-orbit, providing (non-exhaustive list):

- Foreseen operational modes and resources consumption.
- Description of activities performed and duration.
- Description of needed services and environmental constraints.
- Identification of degraded cases.

5.1.1 Payload Categories

The payload design activity should also take into consideration to identify a payload category in terms of two different point of views, payload needs and payload mobility.

Talking about payload needs, we can have:

- **Standard Payloads (STD):** payloads that do not require late access/early recovery nor field of view or need for direct space exposure.
- **Late-Access (LA) / Early-Retrieval (ER) Payloads:** payloads that require to be installed just prior to launch and/or to be recovered just after landing.
- **Field-of-View (FoV) / Direct Exposure Required (DER) Payloads:** payloads that require to be directly exposed to the space environment and/or to perform observation pointing out of MPCB.

In terms of payload mobility instead, an additional classification can be made:

- **Fixed Payload (F-PL):** any Payload which does not separate from Space Rider MPCB and remain fixed in its MPCB compartment.
- **Deployable Payload (D-PL):** any Payload, which can separate from Space Rider MPCB into its own free-flying mission, divided in three sub-classes:
 - Payload deployable with no manoeuvre capability (**D-PL (NM)**)
 - Payload deployable with manoeuvre capability (**D-PL (M)**)
 - Payload deployable with operations within the Space Rider Keep Out Zone (retrieval/re-visitation) (**D-PL (KZ)**)
- **Movable Payload (M-PL):** any Payload which does not separate from Space Rider MPCB but perform movement inside it (i.e., robotic arm).

5.2 Safety requirements

Regarding the applicable safety requirements to P/L design, development and verification process please consider that proper applicable documentation will be provided (after formal contact). This documentation includes the technical and system safety, space debris and collision avoidance requirements applicable to SRS payloads (including payload-provided ground and flight support systems) to be followed by the Customers and Experiment Owners (see [RD3]). These requirements are applicable to the ground and flight operations, from the launch preparation activities of the flight vehicle to the recovery activities of the payloads after landing. The Customer shall interface with the Experiment Owners to assess the implementation of the safety requirements at payload level. The process to reach the Safety Acceptance of the MPCB payloads and experiments is described in [RD4]. During the P/L design, development and verification process Customers should take into account the following aspects related to safety (non-exhaustive list):

- **Severity classes:** the consequences of failure of one of the elements of the P/L are prioritized in severity classes (definitions are reported in [RD4]).
- **Hazard prevention, control, and minimization:** the risks and related consequences must be clearly identified and included in the P/L safety documentation. Where applicable an adequate number of safety barriers shall be provided. During ground operations the P/L functions which may lead to specific consequences should ensure a safety interception capability, by providing the status and the ability to command or inhibit at least one of the barriers. In addition, differently from a standard mission, the Space Rider System will include operation at landing site after the orbital mission and related safety aspects.
- **Materials:** safety requirements and aspects related to the presence of hazardous and forbidden materials (chemical releases, flammable materials, organic and contagious products, genetically modified organisms, etc.)
- **Fluid systems:** safety requirements and aspects related to the presence of this kind of systems (leakage, pressure, etc.)

- **Combustion, high-temperature, and heat-exchanger systems:** safety requirements and aspects related to the presence of this kind of systems (skin-contact safety threshold, potentially noxious exhausts, gaseous pressure, etc.).
- **Batteries, static electricity, and electrical systems:** safety requirements and aspects related to the presence of this kind of systems (isolation, grounding, leakage, fire, etc.).
- **Radiation systems (non-ionising, ionising, optical instruments, and LASERs, etc.):** safety requirements and aspects related to the presence of this kind of systems (protections, Threshold Limit Values, Biological Exposure Indices, etc.).
- **Pyrotechnic systems:** safety requirements and aspects related to the presence of this kind of systems (failure propagation, generation of space debris, etc.).
- **Space-debris mitigation and collision avoidance:** safety requirements and aspects related to the possibility to generate space-debris or in-flight collisions.

5.3 Product Assurance Requirements

ESA recommends to follow the ECSS standards (at latest issues) with availability to support Customer to address the applicable and required standards.

5.4 Cleanliness and Contamination Requirements

The P/L FM item, in addition to selected test models must comply with SR cleanliness and contamination requirements (see section 4.5). If the P/L presents any deviations to the above ESA shall be informed.

As part of hand-over / incoming inspection activities, cleanliness measurements will be made to assess the cleanliness levels of the P/L.

5.5 Design Reviews and Follow-Up

In order to monitor the P/L design, development and verification process and the expected output through time, the MPCB Operator (as Payload Aggregate responsible) will request the Customer to certify, at certain point during flight preparation (see ANNEX B), the P/L design maturity level according to the following criteria:

- Payload performances: for information.
- Payload external interfaces: for approval.

Using the recommended definitions and the typical ECSS project life cycle described in [RD8] the expected reviews (or equivalent level maturity key point, see [RD7]) to reach the above criteria are:

- Preliminary Design Review (PDR)
- Critical Design Review (CDR)
- Qualification Review (QR) / Acceptance Review (AR)
- Pre-Shipment Review (PSR).

Relevant milestones must be achieved and certified (Certificate of Compliance) by the provision of proper documentation to MPCB Operator or equivalent evidence of comparable

technical achievements in support to the required certification shall be produced by the Customer.

The MPCB Operator is available to participate as observer during Customer identified P/L main reviews with the aim to support the Customer and verify the P/L interface requirements compliance with respect to MPCB interface point of view.

In case a Customer express the need, a P/L design and development process follow-up can be provided by ESA on-demand. The purpose of this follow-up activity is to be intent not for qualification purposes but as knowledge support for the maturity reaching point.

5.6 Payload Deliverables

Customer are required to provide a set of deliverables during the preparation to flight in accordance with the approach described in section 5.5. It will be considered as deliverable the P/L technical documentation with impacts on SR MPCB interfaces, contractual, and safety-related documentation plus possible required P/L test models (see section 5.6.4).

5.6.1 *Payload Design and Development Technical Documents*

In this section an overview of the principal P/L design and development technical documentation with impacts on SR MPCB interfaces will be presented. The customer shall collect all the needed technical information for the P/L acceptance process, e.g. (non-exhaustive list).:

- P/L Interface Requirements Document (P/L IRD)
- P/L Ground and Flight Operations Plan
- P/L User Manual
- Qualification / Acceptance Test Report
- Certificate of Conformity (CoC)

The full list of expected deliverable technical documentation will be formally listed in the Statement of Work (section 5.6.2) in accordance with the approach described in section 5.5.

5.6.2 *Payload Contractual Documents*

An overview of the contractual documentation that will be required to be formally stated between the P/L Customer and the MPCB Operator to ensure a proper legal framework between the two entities.

Memorandum of Understandings (MoU)

The document where it is formally stated between the P/L Customer and the MPCB Operator the commitment to continue the work and mutually commit for Customer P/L flight with one of the SRS planned flights.

Flight Service Agreement (FSA)

The document where it is formally stated between the P/L Customer and the MPCB Operator the commitment to continue the work and mutually commit for Customer P/L on a specific and identified SRS flight (in particular on the Maiden Flight). Applicable to the FSA are the following programmatical and technical documents:

Statement of Work (SoW)

The document where it is formally stated between the P/L Customer and the MPCB Operator commitment to the list of activities both P/L Customer and MPCB Operator side to be performed to accept P/L aboard SR according to the required technical documentation issued during Payload Aggregate preparation process.

Interface Control Document (ICD)

The document deals with the technical definition and control of the applicable interfaces between the SRS and each P/L Customer part of the MPCB Payloads Aggregate for a specific flight. The scope of this document is to assure and maintain compatibility and coherence of interfaces development and implementation within the SRS mission and each embarked Payload. The ICD will be kept up to date according to configuration evolution as defined in the Payload Aggregate preparation process.

5.6.3 Safety Documents

The documentation expected along the P/L Safety Acceptance Process is reported in [RD4]. The Customer shall provide to the MPCB Operator a preliminary safety analysis for the P/L to identify the potential risks with consequences on the safety of people, goods and the environment that could be generated by the P/Ls, and their mitigation measures.

5.6.4 Payload Models

As a verification method for P/Ls, an optimized number of representative P/L model's (both analysis and physical models) are identified to achieve a high confidence level in the P/L verification with the shortest schedule and a suitable weigh of costs and risks. The P/Ls model's delivery (see ANNEX B) is planned to decouple the availability vs possible launch date evolutions and the verification sequence is also structured to synchronize with the SRS qualification program.

Note that if the Customer is able to provide the P/L FM at expected aggregate integration tests (see section 6.4) requirement to provide specific P/L representative models can be amended.

Payloads Analytical Models²²

Preliminary accommodation and mission feasibility studies relies on P/L analytical models that will represent in a numerical domain some of the main characteristics of the P/Ls.

Note that the thermo-mechanical verification of the aggregate is performed via coupled load analyses, so stringent requirements to P/L mathematical models' correlation and accuracy are enforced to P/L to ensure quality of analysis results. The development of P/L FEM and TMM models must follow correlation criteria that will be provided as applicable technical notes.

Computer-Aided Design Models (CAD)

A digital 3D model representing the P/L. Information to be reported are sizes, shape, interface's location, and mass to perform a digital fit-check and accommodation analysis.

Finite Element Model (FEM)

A digital approximation of the P/L structure into simpler pieces whose behaviour can be described by equations. Used to perform structural analysis, mass properties, and preliminary and final mission analysis.

²² Additional analytical models (e.g., dynamic envelopes, ...) can be required for specific P/L types (e.g., robotic arms).

Thermal-Mathematical Model (TMM)

A digital numerical model of the P/L represented by concentrated thermal capacitance nodes or elements, coupled by a network made of thermal nodes. Used to perform a thermal analysis to size the design and to support the thermal testing verification.

Payloads Physical Models

High Fidelity 3D Mock-Up (3D Mock-Up)

A 3D model representative of P/L shape to realize the necessary fit checks during verification phase. The model must be an accurate and precise dimensional reproduction of the P/L external shape, representative of mechanical I/F, provide location of the grounding, electrical, data and any other applicable I/F (e.g., fixation points to interface with MPCB thermal blanket). Late-Access P/Ls model shall also allow preliminary check for the late installation operations therefore any doors or drawers to be opened must be implemented. No constraints are applicable to model material, fit checks and integration tests will be done inside the RM FM Multi-Purpose Cargo Bay, so it is required the compliance of the P/Ls model with the cleaning requirements.

Electrical and Functional Model (EFM)

A model representative of the P/L data and power connector pin function, as well as representative of the real unit from a functional behaviour and data interface point of view to realize the preliminary electrical and functional tests. The model must be representative of P/L electrical and data interfaces and P/L representative software. A lower standard for electrical components, materials and processes may be used as long as they provide representative performances but in compliance with the cleaning requirements.

Representative Dummy (Dummy)

A model mechanically representative of the characteristics of the P/L in order to be able to implements the fit checks necessary in verification phase. It shall be accurate and precise in

mass and dimensional representativity of external shape of the FM item with similar superficial treatment. The model must be representative of mass, CoG, momentums of inertia, mechanical I/Fs necessary for integration of the payload any other applicable I/F (e.g., fixation points to interface with MPCB thermal blanket). In case of non-availability of the P/L Flight Model at required time, the dummy model will be used as back-up and then would fly. Thus, the FM requirements are made applicable to it, therefore it shall comply with product assurance and all cleanliness and contamination requirements. A single instance will be required contractually to mitigate the risk of P/L FM unavailability at the required delivery date for final integration.

Flight Model (FM)

The qualified and flyable model of the P/L needed for final launch campaign integration and checks.

5.7 Export Control

Payload Customers will be requested to provide the classification of Payload and its components in terms of export control regulations.

6 AGGREGATE VERIFICATION AND INTEGRATION

6.1 The Payload Aggregate

The selected P/Ls to be hosted in the MPCB for a specific SR mission are identified as the Payload Aggregate (PLAG). This configuration is composed by a group of Customer's developed and qualified P/Ls to be accommodated in the MPCB in a specific configuration defined by the Aggregate Design Authority (ADA). The aggregate, managed by the MPCB Operator, will be considered as a product, and will pass through a preparation, verification, and integration process to be qualified for the flight.

6.2 Aggregate Preparation Process

The aggregate preparation is an end-to-end process that will identify and prepare the best configuration of payloads to be flown for a specific SR mission:

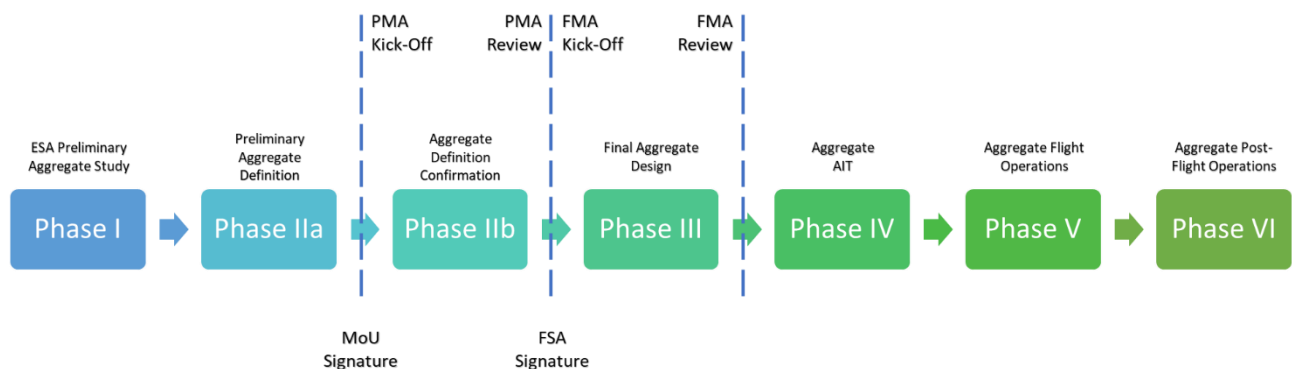


Figure 6-1 Payloads Aggregate Preparation Process.

- **Phase I:** ESA Preliminary Aggregate Study, early Customer contacts, preliminary compatibility evaluation of P/L with SRS.
- **Phase IIa:** Preliminary Aggregate Definition by Aggregate Design Authority (ADA).
 - **MOU signature:** Formal Customer/MPCB Operator commitment to fly on one of six SRS flight.

- **Phase IIb:** Aggregate Definition Confirmation, aggregate design coherent with Preliminary Mission Analysis (PMA)
 - **FSA signature:** Formal Customer/MPCB Operator commitment to fly on one identified SRS flight.
- **Phase III:** Final Aggregate Design, detailed aggregate design coherent with Final Mission Analysis (FMA)
- **Phase IV:** Final Aggregate AIT
- **Phase V:** Aggregate Flight Operations
- **Phase VI:** Aggregate Post-flight Operations

6.2.1 Phase I: Payloads Feasibility Phase

At this phase the Space Rider System Operator collect requests from Customers, via open calls for flight opportunities or direct reception of P/L proposals and evaluate a preliminary compatibility of P/L with SRS. The Customer, with MPCB Operator technical support, will define P/L requirements, check preliminary P/L accommodation, build interface and preliminary contractual documentation. A feasibility study will bring the potential P/Ls to Phase II.

6.2.2 Phase IIa: Preliminary Aggregate Definition

All the P/Ls that are part of the feasibility study are required to provide their preliminary data regarding requirements and interfaces and safety aspects. A preliminary P/L accommodation study is then performed by the Aggregate Design Authority (ADA).

At the end, with the results of the feasibility and accommodation studies a Preliminary Aggregate Definition is generated, and all the P/Ls selected will be invited to sign a Memorandum of Understanding (see 5.6.2).

6.2.3 Phase IIb: Aggregate Definition Confirmation

At this phase all the P/Ls that are part of the preliminary aggregate design are required to provide their updated preliminary data regarding requirements and interfaces, safety aspects and software models. After that Customers are required to demonstrate the P/L maturity at PDR-level (see 5.5) with the MPCB Operator to verify the preliminary architectural design and mission concept of operations. In the meantime, the Aggregate Design Authority (ADA) will use the P/Ls provided data to start a Preliminary Mission Analysis (PMA).

At the end, with the results of the PMA and the P/Ls PDRs there will be the P/Ls selection for a specific mission and signature of relative Flight Service Agreement (see 5.6.2).

6.2.4 Phase III: Final Aggregate Design

At this phase all the P/Ls that are part of the final aggregate design are required to provide their consolidated data regarding requirements and interfaces, safety aspects and analytical models. After that Customers are required to demonstrate the P/L maturity at CDR-level (see 5.5) with the MPCB Operator to verify the final architectural design and mission concept of operations. The Aggregate Design Authority (ADA) will then use the P/Ls provided data to start the Final Mission Analysis (FMA) while the Customers are required to provide P/L's 3D High Fidelity Mock-Ups for fit checks on FM MPCB and Electrical and Functional Models for tests with the RM ATB. At the end of this process, P/Ls are required to sustain or provide certification of compliance for the Qualification Review (QR) and an update of safety information.

At the end, with the results of the FMA and the P/Ls CDRs and QRs there will be the P/Ls confirmation for the specific SR mission.

6.2.5 Phase IV: Final Aggregate AIT

All the P/Ls selected for the flight must provide a P/L Flight Model or at least a Representative Dummy at final P/L integration into RM MPCB at integration site in Europe. Then the RM will be moved to the launch site, at Europe's Spaceport in Kourou, French Guiana.

After final checks the MPCB door of the SR RM will be closed in preparation for the integration with the AOM module. The SRS is now ready for the Flight Readiness Review (FRR). After this

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milestone, the SR Vehicle (RM and AOM) and the fairing are being integrated with the VEGA-C launch vehicle. At this point, at the ELV, the late access window for P/Ls integration starts. The two LA compartments are accessed contemporary by two operators, installing the environment sensible P/Ls into their housings. When the late-access procedure is finished, the system can face the last milestone before the lift-off, the Launch Readiness Review (LRR).

6.2.6 Phase V: Aggregate Flight Operations

This phase nominally begins at launch time, and after the ascent phase and LEOP activities enter the commissioning phase with the initial RM mission attitude and MPCB door aperture. At this point in time begins the P/Ls experiments window with the execution of mission timeline during each 90 minutes orbit at different attitudes. The experiments will undergo for plus than 60 days exchanging of TM&TC from/to VCC and PGCC. After two-months in space the experiments window ends and one day is required to let the avionic and P/Ls cool-down in preparation for the re-entry. The MPCB door will be closed and after the reception of the authorization from Landing Site the SR vehicle will perform a de-orbit burn and the separation of RM (for re-entry) and AOM (for destructive re-entry). The RM will perform the re-entry and descent phase with its sub-sonic parachute followed by the deployment of a guided parafoil to steer the RM vehicle up to its touchdown at Landing Site.

6.2.7 Phase VI: Aggregate Post-flight Operations

At the landing site the SR RM will be cooled-down and powered-down to allow operators access LA compartments to retrieve the environmental sensitive P/Ls. Standard P/Ls are recovered once the RM is moved to the post-landing facility, where the hand-over of P/Ls from MPCB Operator to owners can be performed while the RM is moved to refurbishment phase.

6.3 Aggregate Verification

The aggregate verification process will verify the compliance to the requirements for a specific SR mission and its acceptance. If the results of the verification process are satisfactory, the Payloads Aggregate is ready to perform the mission onboard Space Rider.

6.4 Aggregate Compatibility Analysis and Tests

The following analyses and compatibility tests will be performed for the qualification and acceptance of the overall aggregate (see ANNEX B for further details):

6.4.1 System-Level Mission Analyses

Preliminary Mission Analysis

All the Aggregate Payload's must provide (at PDR level) preliminary data for mechanical aspects, power consumption, thermal dissipation, EMC compatibility, operational aspects, and relative analytical models (see 5.6.4). The ADA will use all the provided inputs to define the PMA to establish the correlation between P/Ls and their mission requirements.

Final Mission Analysis

All the Aggregate Payload's must provide (at CDR level) consolidated data for mechanical aspects, power consumption, thermal dissipation, EMC compatibility, operational aspects, and relative analytical models (see 5.6.4). The ADA will use all the provided inputs to define the final mission analysis to establish the correlation between P/Ls and their mission requirements.

6.4.2 System-Level Fit Checks

All the Aggregate Payload's must provide (at CDR level) a proper high-fidelity 3D mock-up to perform P/L accommodation fit checks on the RM FM.

6.4.3 Preliminary Integration, Electrical and Functional Tests

All the Aggregate Payload's must provide a proper Electrical and Functional Model (see 5.6.4) to perform preliminary integration, electrical and functional tests on the RM ATB.

6.4.4 Final Integration, Electrical and Functional Tests, Mass Properties

All the Aggregate Payload's must provide a proper FM or representative dummy (see 5.6.4) to perform final integration, electrical and functional tests on the RM FM.

6.4.5 Mechanical, Thermal and EMC Analyses for Aggregate Acceptance

All the Aggregate Payload's information provided at integration will be used by the ADA as inputs to define the final mechanical, thermal and EMC analyses for the aggregate acceptance review.

6.5 Aggregate Assembly, Integration and Test

The SR baseline locations for the Payload Aggregate – Assembly, Integration and Test (AIT) activities are:

- Standard Payload AIT Site(s):
 - ESA ESTEC in Noordwijk, The Netherlands (only MF).
 - TAS-I Premises in Turin, Italy (sub-sequent flights).
- Late-Access / Early Retrieval Payloads AIT Site:
 - Centre Spatiale Guyanese in Kourou, French Guyana.

The Standard Payloads AIT operation with RM FM will be performed at AIT site(s) in Europe and back-up activities are based on P/L Representative Dummy. On the contrary operations on Late Access (LA) P/Ls or cartridges will be carried out at launch site where suitable equipment and facilities will be put in place to support the relevant tasks.

Every SR mission can be slightly different in terms of P/L logistic accommodation.

6.6 Aggregate Safety Acceptance Process

Customers are required to provide P/Ls safety-related information as described in section 5.2 and 5.6.3. The MPCB operator, as Aggregate Responsible will collect the information from all aggregate P/Ls and will be subjected to a Payload Aggregate Safety Acceptance Process (managed by the designated Safety Authorities), which includes safety submissions and associated milestones, that are reported and described in detail in [RD4]. This process outputs in terms of safety feedback or required actions related to P/Ls safety aspects will be managed and forwarded to Customer by the MPCB Operator.

7 SPACE RIDER SERVICES FOR PAYLOADS

Payload services section reports the list of all the capabilities that the SR RM can provide as a service to the hosted payloads. Services can be provided as:

- **built-in capability** of SR system via MPCB interfaces.
- **missioning hardware**: a customized set of hardware provided to meet aggregate-level requirements (e.g., power and data cables, ...)
- **extension-kit**: dedicated hardware to be hosted inside the MPCB, to provide a specific required capability.

7.1 Mechanical services

SR Vehicle baseline mechanical services for P/Ls are provided as a built-in capability through its mechanical interfaces (MPCB Support Plates). Technical information about baseline mechanical aspects and limitations can be found in section 3.2.1.

Additional mechanical services not envisioned for baseline (e.g., special mechanical adapters, special fasteners, locking mechanisms, etc.) can be evaluated and eventually provided via the implementation of dedicated extension-kit(s) upon Customer request.

7.1.1 *Standard Mounting – Accommodation Service*

This service is provided by the ADA in order to identify the optimal allocation of the P/L²³ inside the MPCB (in terms of Support Plate allocation, orientation, interference with other P/Ls, ...) taking into consideration its characteristics and requirements that will not require the use of additional and dedicated HW (i.e. P/L directly mounted on the Support Plate without the need of brackets, spacers, ...).

²³ The P/L must be compliant to the MPCB mechanical interface as specified in 3.2.1.
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7.1.2 Non-Standard Mounting – Missioning Hardware Service

This service is provided by the ADA in order to identify the optimal allocation of the P/L inside the MPCB (in terms of Support Plate allocation, orientation, interference with other P/Ls, ...) taking into consideration its characteristics and requirements that will require the use of additional and dedicated HW.

The requirement for additional and dedicated HW for P/L mounting can be made necessary in case the P/L requires further structural support (for example to achieve visibility requirements), complex instrument shapes or to satisfy other P/L specific needs.

An example of this kind of service is the supplementary (optional) Support Plate that can be placed to the structural-Y part of the MPCB (see Figure 7-1) in order to accommodate P/Ls that require FoV and/or outer space environment exposition and have no or limited thermal requirements.

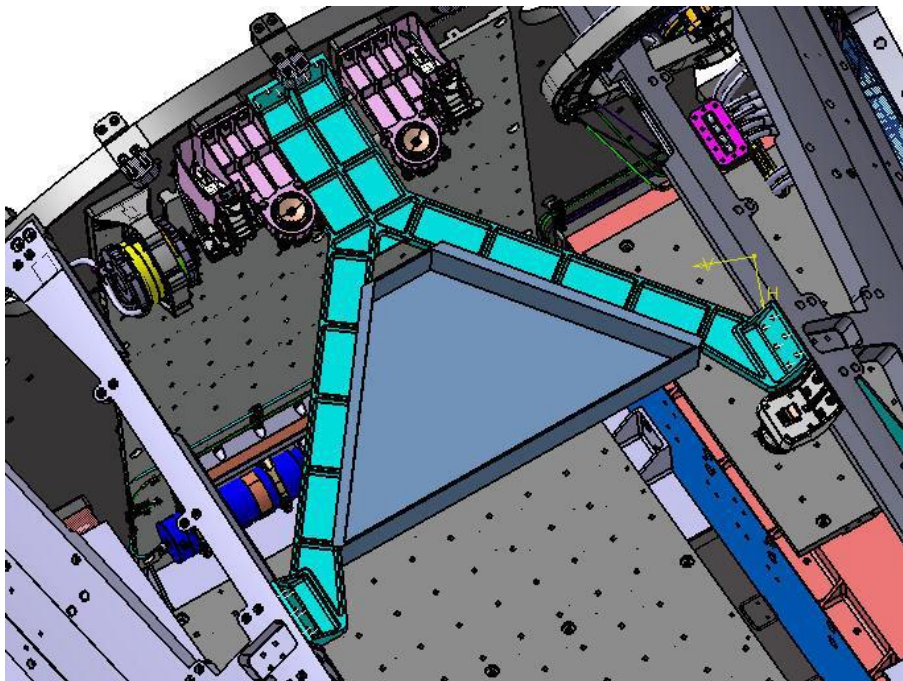


Figure 7-1 Structural-Y optional Support Plate

7.2 Thermal services

SR Vehicle baseline thermal service for P/Ls is provided as a built-in capability through its Thermal Control System (TCS) based on the conductive path to the network of heat pipes provided by the MPCB Support Plates. Technical information about baseline thermal dissipation capability and limitations can be found in section 3.3.1.

Additional thermal services not envisioned for baseline (e.g., high thermal dissipation capability for P/Ls which dissipation through passive thermal control is not feasible or not sufficient) can be evaluated and eventually provided via the implementation of dedicated extension-kit(s) upon Customer request.

7.3 Power services

SR Vehicle baseline power service is provided as a built-in capability through its Power Distribution Unit (PDU) based on several P/Ls dedicated power lines present in the MPCB. Technical information about baseline power supply capability and limitations can be found in section 3.4.1.

Additional power services not envisioned for baseline (e.g., auxiliary-battery, additional PDU, etc.) can be evaluated and eventually provided via the implementation of dedicated extension-kit(s) upon Customer request.

7.4 Data and Communication services

SR Vehicle baseline data and communication services can be divided into In-Flight services, that are services provided to the P/Ls on-board of the vehicle during orbital phase, Ground-based services that are services provided to the Customers via the Ground Segment or Mixed services that are services that involves the mixture of both characteristics.

Additional data and communication services not envisioned for baseline can be evaluated and eventually provided via the implementation of dedicated modifications upon Customer request.

7.4.1 Cybersecurity Aspects

Cybersecurity is a mandatory aspect of space missions, with its importance escalating in tandem with the increasing complexity and connectivity of space vehicles. In this digital era, where cyber threats are becoming more sophisticated and pervasive, the implementation of robust cybersecurity measures is not just a precaution, but a necessity.

By ensuring the protection of critical systems and data, Space Rider cybersecurity measures not only enhance the reliability and safety of its missions but also contribute to the overall success by maintaining the confidentiality, integrity, and availability of vital mission-related information. In this context, Space Rider's cybersecurity framework, with its in-depth security of mission-critical aspects, plays a key role in fortifying the mission against digital threats, thus ensuring its smooth and successful execution.

For missions like Space Rider, effective cybersecurity safeguards are in place to counter a number of potential cyber-attacks that could compromise mission integrity, disrupt communications, or even lead to catastrophic failures.

These security measures are especially significant in the context of Space Rider, where they provide a robust shield against threats like the following:

Interception and Eavesdropping

Transmitted data can be intercepted by unauthorized parties. This data could contain sensitive mission information or control commands. Space Rider ensures that even if the data sent from ground to the orbiter is intercepted, it remains unintelligible and useless to the interceptor.

Unauthorized Command and Control

One of the most critical threats to a satellite is the unauthorized access and issuance of commands. This could lead to a hostile takeover of the satellite's functionalities. Security controls are implemented in Space Rider to ensure that only commands from verified and trusted sources are accepted and executed.

Data Manipulation

Cyber attackers could potentially alter the data being transmitted to the orbiter vehicle, leading to incorrect commands being executed. To counter this Space Rider implements telecommand encryption, combined with authentication, that helps in maintaining the integrity of the data, ensuring that any alterations are easily detectable and preventable.

Man-in-the-Middle Attacks

These attacks occur when an attacker secretly intercepts and possibly alters the communication between two parties. Space Rider make use of robust encryption and authentication protocols make it exceedingly difficult for attackers to insert themselves into the communication channel.

Replay Attacks

This scenario foreseen an attacker that captures a legitimate message and replays it to create an unauthorized effect. The Space Rider implements sophisticated authentication algorithms which includes different factors like time stamps or sequence numbers, which help in identifying and discarding repeated messages.

The benefits of these security measures contributing significantly to raise the resilience of the mission against cyber threats, thereby preserving the integrity and confidentiality of mission-critical data and maintaining the overall safety and success of the mission.

7.4.2 P/L Security in Space Rider: Opportunities for Independent Protection Measures

Space Rider's design includes a transparent and segregated infrastructure for P/Ls, which offers a unique opportunity for Customers to implement their own security measures.

Transparent Infrastructure

The transparency in Space Rider's infrastructure means that payload operators have clear visibility into how their payloads are integrated within the vehicle. This visibility allows payload operators to understand the specific risks and vulnerabilities relevant to their payloads and to design their security measures accordingly.

Segregated Infrastructure

Segregation of the infrastructure ensures that each payload operates within its own dedicated domain, isolated from the others. This segregation is crucial for security, as it prevents any potential cross-contamination of cyber threats between payloads. It also means that security measures implemented by one payload do not inadvertently impact the functionality or security of another.

Opportunities for Payload-Specific Security

Payload operators have the flexibility to implement their own encryption and authentication measures tailored to their specific needs and risk profiles. For instance, a payload requiring high confidentiality can employ advanced encryption algorithms to protect its data. Similarly, payloads with critical control functions can implement robust authentication protocols to ensure that only authorized commands are executed.

This approach allows for a more customized and targeted cybersecurity strategy for each payload, catering to diverse requirements and levels of sensitivity. It empowers payload operators to take charge of their cybersecurity, enabling them to leverage Space Rider's advanced infrastructure while maintaining control over their specific security needs.

In summary, while Space Rider itself may not yet extend its core cybersecurity protections to payloads, its transparent and segregated infrastructure provides a conducive environment for payloads to independently secure themselves, ensuring overall mission integrity and security.

7.4.3 *In-Flight services*

Data-storage service

This service is provided as a built-in capability of the SR Vehicle through its data and communication system allowing P/Ls to store scientific data into RM mass memory. Technical information about data storage capability and limitations can be found in section 3.5.1.

Auxiliary Data Service On-Board

This service is provided as a built-in capability of the SR Vehicle through its data and communication system allowing P/Ls to access a selected set of data of the SR Vehicle on-board, to complement or enhance their own P/L experiment data (e.g., Navigation data, Mission Elapsed Time, Temperature of the I/F plate among others).

7.4.4 *Ground-services*

The P/Ls generated data (Scientific and Housekeeping Data) are downlinked through the Space Rider Ground Station Network to the VCC-OC at Fucino Space Centre, during the orbital phase of the mission. Two types of P/Ls are envisaged:

- **PUS-Compliant P/Ls:** PL HK data is automatically recognized on-board, and it is not stored in the same file as the scientific data. For PUS compliant P/Ls, it is possible to downlink in near real-time the P/L HK data during the visibility windows, if requested. P/L playback HK and scientific data can also be downlinked during the visibility windows, respecting the thresholds reported above for telemetry downlink. Simple TC routing is offered based on APID.
- **Not PUS-Compliant P/Ls:** Real-time downlink is not possible. P/L playback HK and scientific data can be downlinked during the visibility windows, respecting the thresholds reported above for telemetry downlink. Simple TC routing is offered based on P/L ID. It is important that P/Ls that are not PUS terminal clearly distinguish between HK and Scientific files, so that they can be downlinked and processed separately.

If there is only one data source/physical connection for a P/L, it is up to the P/L to discriminate that the files for the different experiments can be uniquely identified (if more than one experiment per P/L).

Flight Monitoring and Commands services

This set of services is provided as a built-in capability of the SRS through the PGCC during mission operations. The PGCC will interface directly only with the Customers P/L through the User Payload Operations Centre (UPOC), which represent the end-users. They will provide all the means for P/L users to both submit user operation requests for P/L planning activities, monitor P/L status and retrieve P/L acquisition data, other than visibility of a subset of pre-agreed information regarding the SR vehicle that can be useful for P/L planning.

PGCC services can be reached from different UPOCs using standard security protocols such as SSL or by the VPN system based OpenVPN/IPSec solution over Internet. Other dedicated links can be supported by PGCC only in case a specific UPOC requires them but in this case the implementation costs will be under the specific UPOC responsibility.

A preliminary list of data exchange from PGCC and UPOC and vice-versa is presented below:

PGCC → UPOC	UPOC → PGCC
P/L live TM (Web-HMI PUS-Compliant P/L)	User Operation Request (Based on the configured templates, via Web-HMI)
Auxiliary live SRS TM (Web-HMI)	P/L Direct Operation Request (PDOR, same as UOR)
Science Raw P/L TM (Query/Download via Web-HMI, this data includes both HK parameters, for PUS-compliant P/Ls, and HK packets)	P/L Configuration Files (Logically part of UOR/PDOR creation)
Playback P/L HK TM	P/L Resource Profiles

(Visualization/Download via Web-HMI, for PUS-Compliant Payload, otherwise may depends on provided pre-processor)	(Required during missionization activities)
Playback SRS Auxiliary TM (Visualization/Download via Web-HMI)	
Mission Planning Data (Web-HMI)	
UPOC Notification (Web-HMI/Offline Email)	
P/L Data Report (Downloadable via Web-HMI/Offline Email)	

Table 7-1: PGCC vs UPOC data exchange list.

Through the PGCC, the ground segment makes available to the end-users a set of services, that will allow an easy management of the P/L requests, planning, monitoring of the P/L activities and data delivery, in particular:

- **Planning Service:** present the status of the mission timeline, together with the data that the end-user has to take into account to request/perform nominal P/L planning. The result of the planning process will be visible on the timeline application.
- **Monitoring Application Service:** It will allow to monitor²⁴ and visualize the P/L behaviour of both real time data and archived data.
- **Commandability Service:** generate and submit the User Operations Requests (UORs) via the front-end web application of the PLMS (located at PGCC). It also allows to implement and submit a fast request for P/L direct command (PDOR) uplink, for selected, critical, and urgent issues, providing all input needed to complete the request.
- **Data Processing and Archiving Service:** manage the retrieval, processing, archiving and distribution of P/L data. Moreover, the DPAS will also manage Space Rider TM/TC

²⁴ The level of monitoring capabilities strictly depends on PL compliance with PUS services.
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auxiliary data and planning data in order to create a consolidated and persistent archive of linked and correlated data. A long-term archive for the entire set of missions will be made available and maintained from the PGCC. The long-term archive will allow to, among others, support P/L's anomalies' resolution.

The level of pre-processing on P/L received data can vary depending on their compliancy with PUS services, in particular:

- For P/L compliant with PUS standard services the following pre-processing capabilities are envisaged.
 - P/L Packet and Parameters extraction, indexing and metadata enrichment.
 - data unit conversion in common PGCC standard ones.
 - extraction and reconstruction, if needed, of P/Ls science data.
 - automatic monitoring of P/L status and command implementation based on the received HK telemetry and science data.
 - comparison of the actual P/L resource, based on recorded telemetry received from VCC, with the nominal ones.
- In case the embarked P/L requires the definition of “custom” PUS services, for example to wrap TM data in proprietary format, only a general set of statistics information and reporting (e.g., amount of data received, events) can be computed. For the actual P/L data extraction, other than the extraction of “custom” PUS packet data field, the execution of user-provided pre-processor can be envisaged, if any. Such pre-processor shall be validated and made available by the P/L end-user.

Access to the specific data is allowed upon authentication of authorized users.

7.4.5 Mixed services

Telemetry & Telecommands (TM&TC) service

This service shall apply to P/Ls that need to upload and download commands and data (e.g., video, housekeeping, science data) at each or during specific orbital passes.

Downlink capability (TM)

The maximum reachable amount of downlink data (Telemetry) capability that can be received from SRS is 2 GB / day for the whole Payload Aggregate at an average downlink data rate of 2 Mbps.

Telemetry encryption service is not provided by default. Further level of encryption can be implemented by P/L Customers (see section 7.4.1).

Uplink capability (TC)

The minimum ensured amount of uplink data (Telecommands) capability is 1 batch of commands per orbit (600 kb / Orbit) for whole Payload Aggregate, with an uplink data rate of 4 kbps. Baseline Telecommands related services includes on-board pre-programmed TCs in the Mission Timeline (MTL), Direct Ground TC, TC File and On-Board Control Procedure (OBCP) services.

Telecommands encryption service is provided by default up to RM level. Further level of encryption can be implemented by P/L Customers (see section 7.4.1).

Dedicated-Antenna service

SR Vehicle baseline does not provide to the MPCB hosted payloads the capability to send/receive data (TM) and commands (TC) through the direct use of SRS radio-frequency channels. All the information from/to P/L must be forwarded using the on-board MMU and the SR GS.

A Dedicated-Antenna service shall apply to P/Ls that need their RF antennas to be installed within the MPCB (with a proper FoV) to acquire and transmit data independently as part of their mission²⁵. This service availability will be evaluated on a case-by-case basis according to SR Vehicle project and mission optimization principles and security verification outcomes.

²⁵ In this case, it is the P/L owner's responsibility to issue their own Request for Frequency Allocation (RFA).
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7.5 Microgravity services

SR Vehicle baseline microgravity services are provided as a built-in capability through its navigation and attitude control system. SR Vehicle can freely drift as it moves along its orbital path with expected g-levels down to 5×10^{-6} g. The maximum disturbances the Payload shall generate towards the re-entry module system is described in section 4.1.8.

7.5.1 *Standard microgravity service*

This service is provided as a built-in capability of the SR Vehicle when P/Ls require to be exposed to standard micro-gravity levels (residual acceleration around 5×10^{-5} g) reachable through different SR vehicle attitudes (see 1.5.3.1). A specific mission and operational timeline considering the defined attitude and duration in accordance with other P/Ls needs and vehicle constraints (e.g., thermal control, etc.) must be evaluated by the ADA during mission analyses.

7.5.2 *Extreme microgravity service*

This service is provided as a built-in capability of the SR Vehicle when a P/L requires to be exposed to extreme micro-gravity levels (residual acceleration greater than 5×10^{-6} g) reachable through a specific SR vehicle attitude (Nose-to-Nadir only, see 1.5.3.1). A specific mission and operational timeline considering the specific attitude and duration in accordance with other P/Ls needs and thermal controls must be evaluated by the ADA during mission analyses.

7.6 Observation / Field-of-View services

SR Vehicle baseline observation services are provided as a built-in capability through its navigation and attitude control system. SR Vehicle can provide pointing capability for observation and/or communication experiments with an angle of view of 90° through the opening the MPCB door. Possible additional structures can be required to integrate the instrument at an adequate position and ensure the wider angle of view (see section 0).

Additional observation services not envisioned for baseline (e.g., fine-pointing, etc.) can be evaluated and eventually provided via the implementation of dedicated extension-kit(s) upon Customer request.

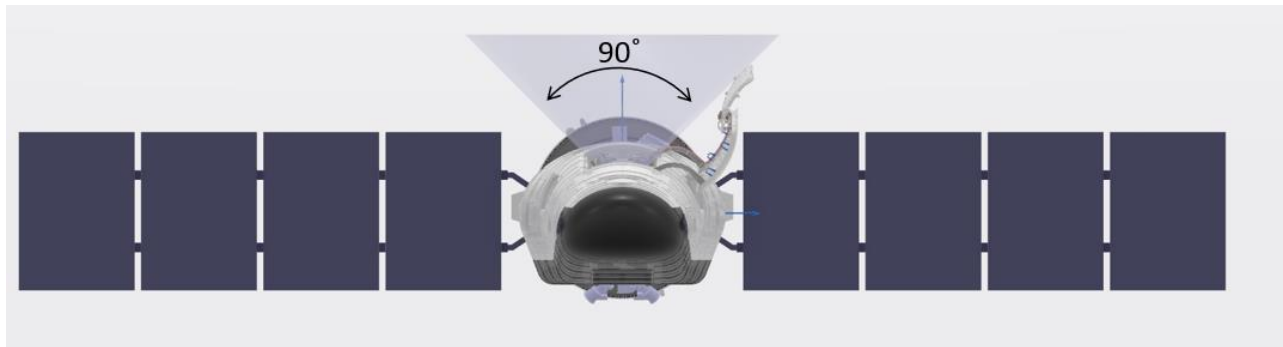


Figure 7-2 SR MPCB Field-of-View capability.

7.6.1 Earth-observation service

This service is provided as a built-in capability of the SR Vehicle when a P/L requires to be pointed directly to Earth through MPCB door upon suitable vehicle attitude placement for specific observation purposes. A specific mission and operational timeline considering the defined attitude and duration in accordance with other P/Ls needs (e.g., PA requirements, etc.) and vehicle constraints (e.g., thermal control, etc.) must be evaluated by the ADA during mission analyses.

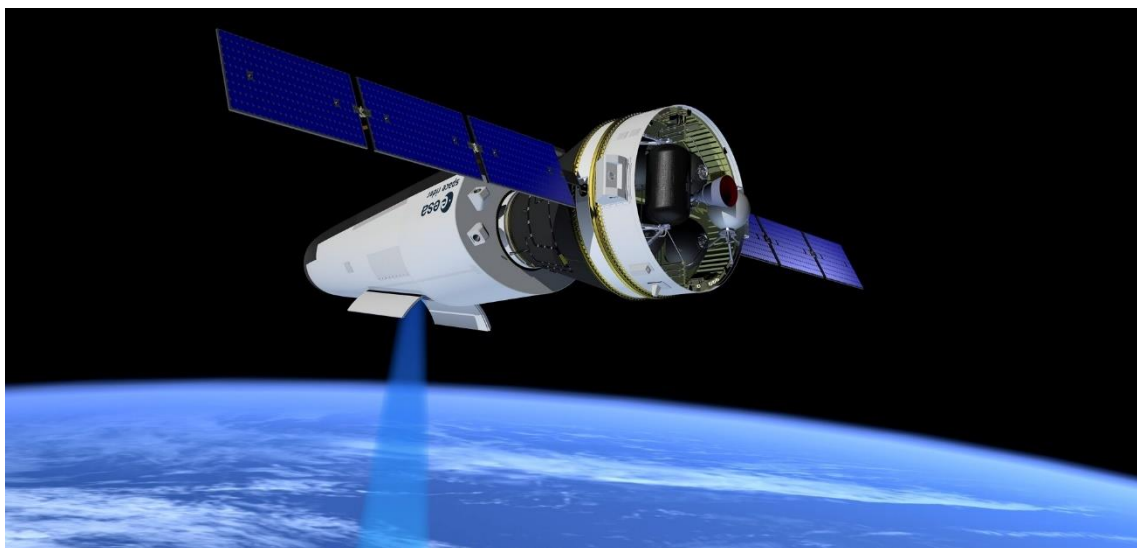


Figure 7-3: SRS during Earth-observation service.

7.6.2 Deep-space observation service

This service is provided as a built-in capability of the SR Vehicle when a P/L requires to be pointed directly to deep space through MPCB door upon suitable vehicle attitude placement for specific observation purposes. A specific mission and operational timeline considering the defined attitude and duration in accordance with other P/Ls needs and vehicle constraints (e.g., thermal control, etc.) must be evaluated by the ADA during mission analyses.

7.7 Exposure / Protection To / From External (Space) Environment

Upon MPCB door opening, the SR enables the exposure of the payloads to the space environment (e.g., radiation, sunlight, ...) as described in chapter 0.

7.7.1 Exposure to external environment service

This service is provided as a built-in capability of the SR Vehicle through the accommodation of the P/L that need exposure to radiation environment for the scope of their flight operations in a proper location inside the MPCB result of the ADA accommodation activities.

7.7.2 Sun-pointing and Sun-avoidance services

Customer's P/L can require the availability of pointing their observation instruments directly to the Sun or total avoiding direct exposure to it.

Sun-pointing service

This service is provided as a built-in capability of the SR Vehicle when a P/L requires to be pointed directly at the sun for specific observation purposes. A specific mission and operational timeline considering the defined attitude and duration in accordance with other P/Ls needs (e.g., PA requirements, etc.) and vehicle constraints (e.g., thermal control, etc.) must be evaluated by the ADA during mission analyses.

Sun-avoidance service

This service is provided as a built-in capability of the SR Vehicle when a P/L requires to avoid direct sun-light exposure or in case be deactivated if the sun enters in their field of view. A specific mission and operational timeline considering the defined attitude and duration in accordance with other P/Ls needs (e.g., PA requirements, etc.) and vehicle constraints (e.g., thermal control, etc.) must be evaluated by the ADA during mission analyses.

7.8 Atmospheric services

SR Vehicle baseline atmospheric services are not provided as a built-in capability. During flight the MPCB hosted P/Ls are exposed to atmospheric / pressure environment described in 4.1.7. Additional atmospheric services not envisioned for baseline (e.g., pressurization, ventilation, auxiliary gases, etc.) can be evaluated and eventually provided via the implementation of dedicated extension-kit(s) upon Customer request.

7.9 Separation, Retrieval and Robotic Handling services

SR Vehicle baseline separation service is provided as a built-in capability through its navigation and attitude control system. SR Vehicle can provide specific attitudes during the release phase of a deployable P/L from its MPCB (e.g., CubeSat separation from its dispenser) into its own free-flying mission. Possible additional structures can be required to integrate the P/L at an adequate position and ensure the wider angle for departure corridor (see section 0).

Additional contextual services not envisioned for baseline (e.g., retrieval service, robotic handling etc.) can be evaluated and eventually provided via the implementation of dedicated extension-kit(s) upon Customer request.

7.10 Additional services (Upon Customer's Request)

Services or capabilities not listed above but considered required by a Customer can be evaluated and estimated for implementation upon specific Customer request.

8 DOCUMENTS

8.1 Reference Documents

Users can refer to the following documents for reference information at their latest available versions:

Ref Id	Doc Id	Title
RD1	N/A	VEGA User's Manual
RD2	N/A	VEGA-C User's Manual
RD3	ESA-ST-SR-ST-2022-0001	Space Rider System – Payload Safety, Space Debris and Collision Avoidance Requirements
RD4	ESA-ST-SR-PL-2022-0001	Safety Process for Space Rider Re-Entry Module Payloads
RD5	ECSS-E-ST-32-10C	Structural factors of safety for spaceflight hardware
RD6	ECSS-E-ST-70-41C	Telemetry and telecommand packet utilization
RD7	ECSS-M-ST-10-01C	Organization and conduct of reviews
RD8	ECSS-M-ST-10C	Project planning and implementation
RD9	N/A	RM Simplified 3D CAD Model

Table 8-1: Reference documents.

9 ACRONYMS

AAR	Aggregate Acceptance Review
ADA	Aggregate Design Authority
AIT	Assembly, Integration and Test
ALEK	AVUM Life-Extension Kit
ALTEC	Altec S.p.A.
AO	Atoms of Oxygen
AOM	AVUM Orbital Module
APID	Application ID
AR	Acceptance Review
ATB	Avionic Test Bench
ATOX	Atomic Oxygen
AVIO	Avio S.p.A.
AZ	Approach Zone
B2E	Bay-to-Earth
B2Z	Bay-to-Zenith
CAD	Computer-Aided Design
CoG	Centre of Gravity
CSG	Centre Spatiale Guyanese
CPO	Close Proximity Operations
DLL	Design Limit Loads
D-PL	Deployable Payload
D-PL (KZ)	Deployable Payload with operations in Keep-Out Zone
D-PL (M)	Deployable Payload with manoeuvre capability
D-PL (NM)	Deployable Payload with no manoeuvre capability
EA	Early-Retrieval
EAR	Export Administration Regulations
ECSS	European Cooperation for Space Standardization

EFM	Electrical and Functional Model
EGSE	Electrical GSE
EMC	Electro-Magnetic Compatibility
EPS	Electrical Power System
ESA	European Space Agency
ESD	Electro-Static Discharge
ESTACK	European Space Tracking
FEM	Finite Element Model
FM	Flight Model
FMA	Final Mission Analysis
FoV	Field of View
F-PL	Fixed Payload
FS	Flight Segment
FSA	Flight Service Agreement
GNSS	Global Navigation Satellite System
GS	Ground Segment
GSE	Ground Support Equipment
GSN	Ground Station Network
H0	Launch date / time
HMI	Human Machine Interface
HK	House Keeping
ID	Identification
IMU	Inertial Measurement Unit
IOD	In-Orbit Demonstration
IOS	In-Orbit Services
IOV	In-Orbit Validation
IP	In-Plane or Internet Protocol
IPSec	Internet Protocol Security
ITAR	International Traffic in Arms Regulations

IXV	Intermediate eXperimental Vehicle
KOZ	Keep-out Zone
LA	Late-Access
LCL	Latching Current Limiter
LEO	Low Earth Orbit
LEOP	Launch and Early Operation Phase
LFL	Low-Frequency Loads
LDS	Landing Site
LS	Launch Site
LSDO	Landing Site Director of Operations
MCC	Mission Control Centre
MCI	Mechanical Capture Interface
MF	Maiden Flight
MGSE	Mechanical GSE
MH	Mobile Hangar
MLI	Multi-Layer Insulation
MMU	Mass Memory Unit
MOC	Molecular Contamination
MoU	Memorandum of Understanding
MPCB	Multi-Purpose Cargo Bay
MTQ	Magneto TorQuers
M-PL	Movable Payload
N2N	Nose-to-Nadir
OBC	On-Board Computer
OBCP	On-Board Control Procedure
OOP	Out-of-Plane
OPSIM	Operational SIMulator
OSZ	Off-Set Bay-to-Zenith
PA	Payload Aggregate

PAC	Particulate Contamination
PDF	Portable Document Format
PDOR	Payload Direct Operation Request
PDU	Power Distribution Unit
PFM	Proto-Flight Model
PGCC	Payloads Ground Control Centre
P/L	Payload
PLAG	Payload Aggregate
PLGSO	Post-Landing Ground Safety Office
PMA	Preliminary Mission Analysis
PPE	Personal Protective Equipment
PPF	Payload Processing Facility
PUS	Packet Utilisation Standard
QR	Qualification Review
QSL	Quasi-Static Loads
RADAR	RAdio Detection And Ranging
REI	Regulations concerning the use of facilities at the CSG
RH	Relative Humidity
RM	Re-entry Module
ROZ	Roll-Out Zone
RW	Reaction Wheels
SPG	Single-Point of Grounding
SR	Space Rider
SRS	Space Rider System
SSL	Secure Socket Layer
SSO	Sun-Synchronous Orbit
STD	Standard Payload
T2S	Tail-to-Sun
TAS-I	Thales Alenia Space – Italy S.p.A.

TC	Telecommand
TCS	Thermal Control System
TM	Telemetry
TM&TC	Telemetry & Telecommand
TMM	Thermal-Mathematical Model
TPZ	Telespazio S.p.A.
TRP	Temperature Reference Point (or Thermal Reference Point)
TZ	Touch-down Zone
UDP	User Datagram Protocol
UOR	User Operation Request
UPOC	User's Payload Control Centre
VCC	Vehicle Control Centre
VCC-LC	VCC-Landing Control
VCC-OC	VCC-Orbital Control
VEGA	Vettore Europeo di Generazione Avanzata
VEGA-C	VEGA Consolidated
VPN	Virtual Private Network

ANNEX A

Possible Applications for Science and Technology Research

Below it will be shown non-exhaustive list of the Scientific and Technology Research fields that may benefit from the Space Rider services.

I. Life Science

- Cell Culture
 - Regenerative Medicine – Organ Transplant
 - Regenerative Medicine – Microorganisms
 - Regenerative Medicine – Stem Cells
 - Stem Cell Biology
 - Tissue Chips
 - Accelerated Disease Models – Cell Culture
 - Monoclonal Antibodies (mAbs) – Cell Culture
 - Vaccine Production – Cell Culture
 - Infectious Disease
- Large Molecule Crystallization
 - MCG
 - mAbs – Crystallization
 - Vaccine Production – Crystallization
- Microorganisms
 - Microorganisms
 - Secondary Metabolites
 - Plant Microbes
 - Biofilms / Corrosion
- 3D Bioprinting
- Accelerated Disease Models – Rodents
- Medical Devices

II. Physical Science Manufacturing

- Deposition
 - Electrodeposition
 - Laser Annealing
 - CVD
 - MBE
 - ALD
 - Spraying and Coating
- Formulation
 - Mixing/Emulsifying
 - Lyophilization
 - Polymerization
 - Flow Chemistry
- Furnace Crystallization
- 3D Printing
- Metal combustion

III. Physical Science R&D

- Shelf life
- Fluid Dynamics
- Thermal Transport
- Interfacial Studies
- Separations

IV. Remote Sensing

- Visible
- NIR/SWIR
- Thermal IR
- RADAR
- RF

V. In Orbit Assembly and Service

- Satellite Deployment
- In Orbit Assembly
- In Orbit Services

VI. Technology Development and TRL



ANNEX B

Space Rider Payload Mission Timeline

Time	Actor	Activity	Type	Milestone
Phase I – Preliminary Aggregate Study				
Commercial opportunities at this stage are evaluated continuously during SR program execution.	MPCB Operator	Open Call for Flight Opportunities.		
	Customer	Response to Flight Opportunities or direct contact for P/L preliminary feasibility evaluation: - P/L Questionnaire delivery.	Document(s) Delivery	
	Customer, MPCB Operator & ADA	Preliminary technical meetings for feasibility study.	Technical meeting(s)	
	MPCB Operator & ADA	First feasibility evaluation.		
Phase IIa: Preliminary Aggregate Definition				
Commercial opportunities at this stage are evaluated continuously during SR program execution.	Customer	P/L IRD and Preliminary Safety File delivery.	Document(s) Delivery	
	Customer, MPCB Operator & ADA	Interface Progress Meetings.	Technical meeting(s)	
	ADA	Preliminary P/L Accommodation Feasibility		
	MPCB Operator & ADA	P/L Selection for Preliminary Aggregate Definition		
	MPCB Operator & ADA	Preliminary Aggregate Validation		Preliminary Aggregate Validation
	Customer & MPCB Operator	Signature of Memorandum of Understanding (MoU)		MoU
Phase IIb: Aggregate Definition Confirmation				

	Customer & MPCB Operator as observer	P/L Preliminary Design Review (PDR) or equivalent		PDR
	Customer	P/L Preliminary Data Delivery: - Documents at PDR Level - P/L Safety File (Phase 0)	Document(s) Delivery	
	Customer	P/L Preliminary Models: - CAD Model - Thermal Mathematical Model (TMM) - Finite Element Models (FEMs)	Analytical Model(s) Delivery	
	ADA	Start of Preliminary Mission Analysis		PMA Kick-Off
	ADA & MPCB Operator	End of Preliminary Mission Analysis		PMA Review
	MPCB Operator	P/L Selection for a specific SR mission - Statement of Work (SoW) to P/L - Interface Control Document (ICD) - Issue 1	Document(s) Delivery	
	Customer & MPCB Operator	Signature of Flight Service Agreement (FSA)		FSA
Phase III: Final Aggregate Desing				
	Customer & MPCB Operator as observer	P/L Critical Design Review (CDR) or equivalent		CDR
	Customer	P/L Consolidated Data Delivery: - Documents at CDR Level (as specified in the SoW) - P/L Safety File (Phase 1)	Document(s) Delivery	
	Customer	P/L Consolidated Models: - CAD Model - Thermal Mathematical Model (TMM) - Finite Element Models (FEMs)	Analytical Model(s) Delivery	
	Customer	High-Fidelity 3D Mock-Up delivery.	Physical Model(s) Delivery	
	SRS Mission Operator	Start of Final Mission Analysis.		FMA Kick-Off
	ADA	P/Ls Fit Check on RM MPCB.		Fit-Checks

T0 – 10 Months	<< <i>Space Rider RM PFM moved to ESTEC for ETC</i> >>			
	Customer	P/L Electrical and Functional Model (EFM) delivery	Physical Model(s) Delivery	
	ADA	P/Ls EFM Model Integration, Electrical and Functional Tests on RM ATB.		Preliminary Electrical and Functional Tests
	Customer & MPCB Operator as observer	P/L Qualification Review (QR) or equivalent		QR/AR
	Customer	Documents post QR (as specified in the SoW) P/L Safety File (Phase 2) delivery	Document(s) Delivery	
	MPCB Operator	Interface Control Document (ICD) - Issue 2	Document(s) Delivery	
	ADA	End of Final Mission Analysis		FMA Review
Phase IV: Final Aggregate AIT				
	Customer	Pre-Shipment Review (PSR)		PSR
	Customer	PSR Documentation (as specified in the SoW)	Document(s) Delivery	
	Customer & MPCB Operator	P/L FM and/or Representative Dummy delivery	Physical Model(s) Delivery	Hand-over to MPCB Operator
	ADA	P/Ls Integration, Electrical and Functional Tests, and Mass Properties		Final Electrical and Functional Tests
T0 – 3 Months	<< <i>Space Rider RM moved to Launch Site at Kourou</i> >>			
T0 – 2 Months	Customer	Environmental sensitive P/L delivery	Physical Model(s) Delivery	Hand-over to MPCB Operator
	Customer	P/L Safety File (Phase 3) delivery	Document(s) Delivery	
	ADA	Aggregate Acceptance Review		AAR
	RM Design Authority	MPCB Main Door Closure		

	SRS Mission Operator	Flight Readiness Review		FRR
T0 – 36 h	<< <i>Space Rider System (SRS) and fairing integrated with VEGA-C Launch Vehicle at Launch Pad</i> >>			
	Launch Operator	Launch Readiness Review		LRR
T0 – 17 h	ADA	Late-Access window START for P/Ls integration		Late-Access Integration Start
T0 – 16.5 h	ADA	Late-Access window END for P/Ls integration		Late-Access Integration End
T0 – 8 h	Launch Operator	Chronology Start		
Phase V: Aggregate Flight Operations				
T0	<< <i>Launch</i> >>			
	Launch Operator	Ascent Phase and LEOP Commissioning and MPCB Main Door opening		
T0 + 3 h	MPCB Operator	Start of Experiments Window		
T0 + 60 days	MPCB Operator	End of Experiments Window		
L0 – 1 day	SRS Mission Operator	Avionic and P/Ls Cool-Down		
	Operations Authority	MPCB Main Door Closure Authorization from Landing Site De-Orbit burn Separation of RM (for re-entry) and AOM (for destructive re-entry).		
	Operations Authority	Coasting, re-entry, and descent. Sub-sonic Parachute deployment (M=0.2) at 6-10Km alt. Guided Parafoil deployment.		
L0	<< <i>Touchdown</i> >>			
Phase VI: Aggregate Post-flight Operations				
	SRS Mission Operator	RM Vehicle Power-down, deactivation of the electric and pyrotechnic circuits (except OBC, PDU and Payload Aggregate powered up at 50W).		
L0 + 2h	SRS Mission Operator	Operators are allowed to perform payloads early retrieval.		

L0 + 2.5h	MPCB Operator & Customer	Early-retrieval Payloads are moved to PPF. Hand-over of P/Ls from MPCB Operator to owners.	P/Ls FM back to Customer	
	SRS Mission Operator	RM Vehicle moved to the decontamination Facility		
	SRS Mission Operator	Operators are allowed to perform MPCB access for retrieval of Standard P/Ls.		
L0 + 16 days	MPCB Operator & Customer	Hand-over of P/Ls from MPCB Operator to owners.	P/Ls FM back to Customer	
	SRS Mission Operator	RM Vehicle prepared for shipment to the refurbishment facilities.		To Refurbishment

Table 0-1: Space Rider Mission Timeline

End of Document